GEATOMICA

|

MATERIALS ASSURANCE

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PUBLIC EVIDENCE EDITION V1.0.1 | OCTOBER 2026

Geatomica

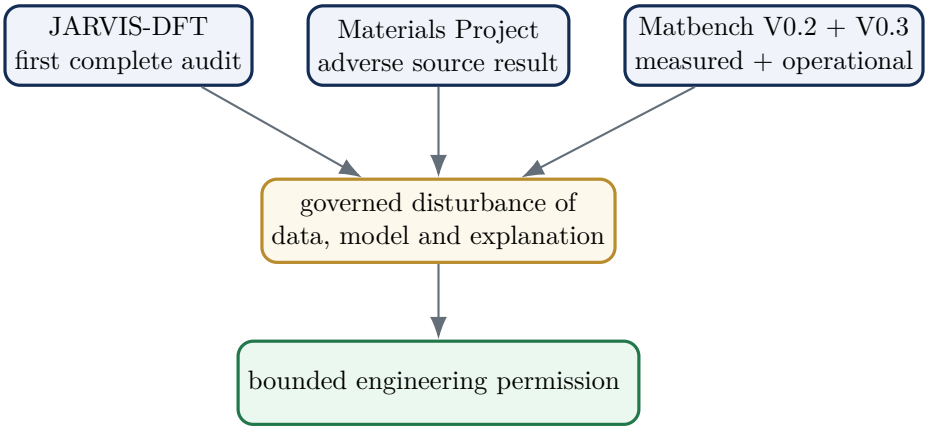
A Materials Assurance Framework for Descriptor Authority, Model Viability and Engineering Trust

A failure-first,
cross-source study of materials data,
models,
explanations
and the decisions built around them

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Scientific evidence state.
This evidence edition integrates:

- a JARVIS-DFT four-target audit and numerically equivalent local replay, with raw-byte divergence preserved;
- a Materials Project programme whose energetic/electronic evidence remains bounded and whose raw elastic interpretation remains on source hold;
- a completed CPU Matbench V0.2 nomination run containing 20 valid official-fold units and paired permutation-target nulls;
- a sealed 3,440-unit Matbench V0.3 fresh assurance run with exact same-environment repeat completed;
- a sealed Spatial V0.5 factorial replay and a final Spatial V0.6.1 confirmation in which 576 fresh and 576 replay units were byte-, metadata- and numerically identical; bulk metric geometry confirmed while shear polarity remained conditional.

The demonstrated scientific evidence is materials-specific; the runtime, replay, memory and authority contracts are formulated independently of materials semantics, making cross-domain transfer a testable successor hypothesis rather than a present validation claim.
The manuscript reports measured, replayed, report-bound and operational evidence as separate classes. It is not peer reviewed, does not authorise material selection or qualification, and does not claim universal descriptor hierarchy, physical causality, experimental validation or external validation.

October 2026

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Corrected preprint successor to archived v1.0.0

Authorship, Evidence and Publication Status

Joseph Maxwell is the sole human author and research lead of the Geatomica research programme. AI and coding systems assisted implementation, critique, evidence organisation and document production under author review. They do not hold scientific or publication authority.

Statements are classified as **MEASURED LOCAL**, **REPLAYED LOCAL**, **HISTORICAL/ REPORT-BOUND**, **READ-ONLY DERIVED**, **INTERPRETATION** or **PLANNED**. A lower class cannot overwrite raw output or a sealed receipt.

Preprint status: final Spatial V0.6.1 replay integrated. This document is not peer reviewed and makes no external-validation claim.

Revision status. Version 1.0.1 is a maintained successor to the deposited v1.0.0 manuscript. It corrects a documentation-only preprocessing statement, narrows one figure caption, distinguishes two differently bounded percentages, and updates repository navigation. No model was rerun and no accepted numerical result, table or protocol was changed for this revision.

Abstract

Materials models can be predictively useful while remaining unsuitable as explanations or engineering authorities. Geatomica separates source admission, absolute model viability, null separation, descriptor-response stability, representation adequacy, replay and promotion authority.

The evidence programme now includes a JARVIS four-target carrier audit with numerically equivalent local replay; an adverse Materials Project elastic admission result; sealed 3,440-unit Matbench V0.3 fresh and repeat runs with 65,740 compared numeric cells exactly equal; and a sealed, byte-identical Spatial GIL 256 replay. Matbench V0.3 retained strong predictive signal across formation energy, band gap, logarithmic bulk modulus and logarithmic shear modulus. Yet fitted permutation rankings and retrained-removal consequences were almost uncorrelated on average (Spearman approximately 0.039), with only 10 percent mean top-10 overlap. Because the protocol lacks sham full-feature refits, this disagreement establishes that fitted importance cannot be treated as necessity but does not uniquely partition redundancy from optimisation variation.

Spatial GIL first tested whether a complete-shell six-bin angle summary preserved the advantage seen in an earlier 64-material split; it did not. A factorial decomposition then isolated continuous metric geometry as the strongest compact block. In the final V0.6.1 confirmation, bulk metric geometry passed continuity and zero chemical-system-overlap views, while shear polarity failed the required cross-view confirmation. Fresh and replay executions each sealed 576 units; all 1,728 corresponding output files and 927,361 numeric cells were exactly identical. Replay therefore preserved both the positive and held decisions.

The current contribution is bounded local computational evidence. It does not establish causal descriptor hierarchy, engineering qualification, physical or external validation, or external replication.

Reader Map

This monograph in plain language. Question: what remains trustworthy when materials-ML evidence is challenged at its source, model, representation, or execution boundary? **What we did:** we tested those boundaries across JARVIS, Materials Project, Matbench, Spatial, and three bounded engineering studies. **What we found:** fitted use, retrained dependence, null separation, absolute prediction, replay, and permission to act are not interchangeable. **Boundary:** the evidence is local and computational; it is not external replication, physical qualification, or universal engineering authority.

Three reading levels. Just want the answer? Read this box, the evidence boxes in Chapters 4–8, and the Conclusion. **Want the experiments?** Read Chapters 3–8. **Want to audit the programme?** Use Appendices A–D, the public evidence package, and the route-specific receipts.

If you have five minutes. Read the Abstract, the five-line evidence boxes at the start of Chapters 4–8, and the Conclusion. They state the question, comparison, result, decision, and boundary for each empirical route.

If you are reviewing the scientific claims. Chapter 3 defines the programme and its decision rules. Chapters 4–8 contain the measured JARVIS, Materials Project, Matbench, and Spatial results. Appendix C records replay and runtime states. Appendix D gives the claim-relevant scientific configurations.

If you are evaluating descriptor interpretation. Start with Chapter 2, then read Chapters 4, 6, 7, and 8. The central distinction is between what a fitted model uses and what the learning problem can replace after removal and retraining.

If you are evaluating model usefulness. Read Chapter 5 first. It shows why beating a target-permutation null did not imply positive held-out predictive performance for the Materials Project elastic route.

If you are evaluating reproducibility. Read Chapters 8 and 12 and Appendix C. Fresh execution, resume, and replay are kept separate. Byte, semantic, and numerical agreement are reported independently where available.

If you are evaluating engineering relevance. Read Chapters 11–13. The furnace, dual-tank, and heat-shield studies are bounded evidence-to-next-test cases, not deployment validation.

If you are evaluating general-engineering transfer. Read the engineering-witness chapter, the runtime chapter, and the separate Technical & Industrial Companion. The engineering cases provide local control-logic witnesses; they are not retrospective full-stack deployments.

If you are evaluating methods. Appendix D consolidates the claim-relevant datasets, representations, splits, preprocessing, models, seeds, nulls, thresholds, and replay contracts. Private reusable orchestration is not required to inspect these scientific configurations.

Programme in one sentence. The work tests whether apparently useful materials-ML evidence remains useful when the source, population, representation, model interpretation, or execution state is challenged, and it refuses to let a weaker result stand in for a stronger one.

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Chapter 1

Introduction

1.1 Materials engineering begins after the event

Materials engineers rarely receive the process they need to
understand.
They receive what remains after the process has already happened.
A polished and etched microstructure after solidification,
heat treatment
and deformation.
A fracture surface after the applied force has gone and the crack has
stopped moving.
A coating after oxidation,
thermal cycling
or local delamination.
A failed component after manufacturing history,
service conditions
and geometry have combined into one final state.
The evidence is real.
It is also incomplete.
A micrograph does not arrive with labels identifying every thermal
hold,
cooling rate,
contaminant,
stress concentration
or local transformation that produced it.
A fracture surface can constrain the loading history without narrating
that history by itself.

The engineer therefore works backwards.

Possible histories are proposed,

then tested against whatever else those histories should have left

behind.

If fatigue dominated,

should repeated growth markings be present?

If thermal shock initiated the crack,

what temperature gradient and local constraint would have been

required?

If processing caused a brittle microstructure,

which phases,

grain morphologies

or residual stresses should accompany it?

The best explanation is not the story that fits one mark most

beautifully.

It is the history that survives the greatest number of independent

constraints.

This reconstructive logic sits beneath much of materials science.

Composition names the ingredients.

Processing determines what those ingredients become.

Structure records part of that history.

Properties describe how the surviving structure responds.

Failure reveals one of the routes by which the available margin was

lost.

Governing rule

A surviving structure is evidence of the process that produced it.
It is not the process in full.

Machine-learning outputs deserve the same discipline.

A fitted model is another surviving structure.

Its final score,

feature ranking

and predictions are the visible endpoint of decisions made earlier:
what source was queried,
which records were admitted,
which variables were available,
how values were transformed,
where the population was split,
which model family was selected,
what objective was optimised
and which failures were allowed to disappear inside an aggregate.
The model output remains real.
The problem begins when the finished structure is mistaken for a
transparent account of the material system itself.

1.2 From prediction to engineering consequence

Materials informatics has made predictive modelling easier to access
and easier to compare.
Large calculated databases expose formation energies,
electronic properties,
structures
and mechanical responses at scales that would be impractical to
reproduce experimentally.
Benchmark suites provide consistent tasks and evaluation procedures.
Composition descriptors,
structural representations,
graph models
and increasingly large learned representations can produce strong
property predictions.
That progress matters.
The engineering risk does not arise because prediction is useless.
It arises because a predictive result can quietly acquire a stronger
meaning than the experiment actually established.
A feature importance ranking may be narrated as a physical mechanism.

A high R^2 may be treated as evidence that the model is safe to use outside the population on which it was tested.

A populated database field may be treated as scientifically admissible because the source itself is respected.

A repeated top descriptor may be assumed to be necessary.

A model that beats a null may be promoted even when its absolute predictions remain unusable.

Each move begins with genuine evidence.

The overreach occurs in the reconstruction built around it.

This matters because a model does not remain inside a notebook.

It can change what an engineer does next.

A promoted descriptor may influence:

- which material variable is measured routinely;
- which high-fidelity calculation is commissioned;
- which alloy or ceramic family is screened first;
- which experiment receives time and budget;
- which physical interpretation enters a design discussion;
- which route is rejected before fabrication;
- which uncertainty is no longer investigated because the model appears settled.

A descriptor therefore gains practical authority when it begins to change decisions.

That authority should not be inherited automatically from predictive performance.

Engineering consequence

The important engineering question is not only:
Which model predicts best?
It is:
Which part of the model's account is reliable enough to justify the next expensive action, and under what conditions should that permission stop?

This is the problem addressed by Geatomica.

It is a materials-assurance problem before it is a software-governance

problem.

The software exists because the scientific question could not be answered reliably without controlling the route that produced the evidence.

1.3 How the research question changed

Geatomica did not begin as a general assurance framework.

It began as a materials-informatics question.

Could a small set of governing descriptors explain enough of a material response to reduce modelling complexity without cutting away the underlying physics?

The initial physical intuition was straightforward.

Macroscopic mechanical properties do not arise independently of

bonding,

electronic state,

energetic organisation,

structure

and packing.

If some descriptors sit upstream of several material responses, then relationships involving those descriptors might persist across changes in model,

population

or representation.

Such persistence could support reduced-complexity modelling, better measurement prioritisation

and clearer escalation to more expensive calculations.

The first obstacle was feature importance.

A fitted model could produce a plausible ranking, but the ranking answered only where that model routed information under one data state.

It did not establish whether the relationship survived another model, another split,

another population
or complete retraining without the feature.
The question changed:

> Can descriptor stability be measured across controlled changes in the modelling process?

The next obstacle was that stability itself remained insufficient.
A descriptor might recur across seeds or perturbations while the
underlying model remained absolutely unusable.
An apparently stable ranking might also emerge from an inadmissible
target population.
A model could preserve ordering while failing a strict replay
contract.
The question changed again:

> When is the evidence surrounding a descriptor strong enough for it to influence an engineering decision?

This sequence produced the concept of
descriptor authority.
Authority is not another name for importance or stability.
It is bounded permission derived from several evidence objects that
must remain distinguishable.
Figure 1.1
summarises how the research programme evolved.

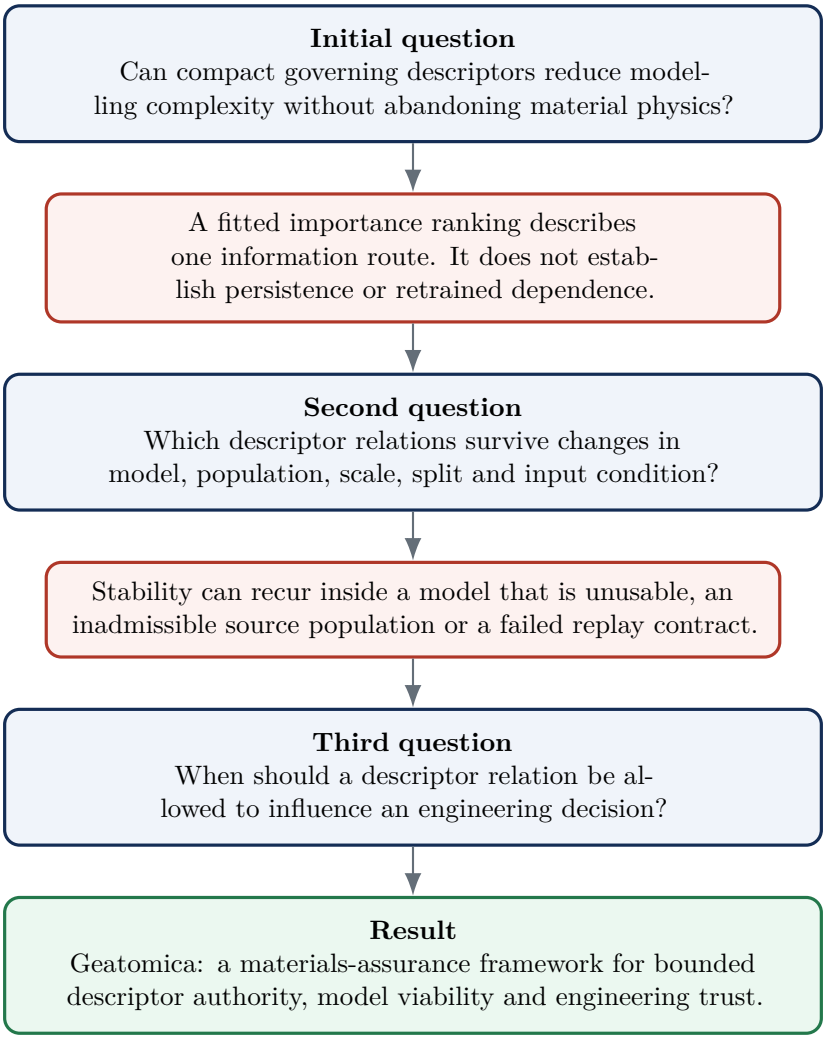


Figure 1.1: Evolution of the Geatomica research question. The framework did not begin as an abstract governance architecture. Each additional layer was introduced because the preceding evidence object proved insufficient for the engineering decision being asked of it.

The framework is therefore best understood as an experimental

consequence.

Evidence admission became necessary because available records were not

automatically interpretable.

Null testing became necessary because predictive performance could be

manufactured by leakage or weak controls.

Absolute viability became necessary because beating a null did not

guarantee useful prediction.

Descriptor disturbance became necessary because one fitted ranking

could not establish persistence.

Retrained removal became necessary because fitted reliance did not establish retrained dependence or physical necessity.

Replay became necessary because one completed execution did not establish reproducibility.

Recovery became necessary because long-running audits could be interrupted without invalidating every completed scientific unit.

Governance became necessary because mixed evidence states could not be represented honestly by one terminal success label.

The architecture grew around the scientific problem.

It should not replace it as the centre of the manuscript.

1.4 The central distinction: importance, stability and authority

The word *important* is routinely used for several different questions.

Geatomica separates at least three.

Table 1.1: The three principal evidence objects separated by Geatomica.

Evidence object	Operational question	What it does not establish
Predictive importance	Where did this fitted model route information under this exact dataset, representation, learner and evaluation state?	Physical necessity, transfer, causality, robustness or engineering permission.
Transformation stability	Does a declared descriptor statistic remain within stated bounds when the allowed modelling conditions are changed?	Absolute model usefulness, source admissibility, causality or truth outside the tested transformation family.
Descriptor authority	Is the combined evidence sufficient to permit use of the descriptor relation for a declared claim or engineering action within an evidenced domain?	Universal validity, material qualification or truth outside that bounded domain.

A descriptor can be highly important and fragile.

It can be weakly ranked but difficult to replace.

It can recur across seeds while collapsing under a chemical-system

holdout.

It can survive within one source while failing in another.

It can remain stable while the carrier model fails absolute viability.

It can appear physically plausible while the source records themselves

are inadmissible.

These are not contradictions to be averaged away.

They are different measurements of how the reconstruction was built.

Why is descriptor authority not simply feature stability plus an applicability domain?

Feature stability,
applicability domains,
uncertainty estimation,
feature importance,
negative controls
and reproducibility each measure part of the problem.
Geatomica's contribution is not to rename those parts.
It is to preserve their distinctions,
measure their disagreement
and prevent one evidence object from impersonating another.
An applicability domain may identify where prediction is reliable
without establishing that the same explanation remains reliable.
A stable feature ranking may recur while the model remains worse than
the mean baseline.
A numerically replayable result may still originate from an
inadmissible source population.
A highly important feature may become replaceable after retraining.
Descriptor authority is therefore an adjudicated relationship among
evidence objects,
not a synonym for any one of them.

This distinction is the centre of the monograph.

The later experiments do not seek one terminal authority score.

They map the boundaries among these evidence objects and preserve

cases where they disagree.

1.5 The empirical programme

The programme developed across three data environments,

each with a different scientific role.

1.5.1 JARVIS-DFT: the first complete audit

JARVIS-DFT provided the first full four-target execution.

It allowed energetic,

electronic

and elastic properties to be studied within a broad calculated

materials population using a frozen common protocol.

The JARVIS stage asked:

- Does enough non-null predictive signal exist to make a descriptor audit meaningful?
- How do descriptor accounts change with target,

  scale,

  model family

  and population split?
- Do the results survive controlled noise,

  dropout,

  shuffle

  and removal?
- Can a separately identified replay reproduce the scientific response surface within a frozen numerical tolerance?

The resulting execution established that the instrument could operate

at meaningful source scale,

return bounded target-dependent behaviour,

survive interruption

and complete a locked replay.

It did not establish external transfer or physical causality.

1.5.2 Materials Project: the adverse source result

The Materials Project stage tested whether the same programme would

survive a second major source environment.

The energetic and electronic routes produced bounded recurrence.

The elastic routes did not produce a simple validation result.

Extreme values,

negative performance

and replay sensitivity exposed a source-admissibility problem that
had not been resolved before fitting.
This was not treated as failed science.
It was treated as a more informative outcome:
the instrument encountered a populated target field,
identified that the available values could not yet support a physical
interpretation,
and refused promotion.
The same programme exposed the difference between relative improvement
and absolute viability.
This result produced the two-door model-viability rule developed later
in the monograph.

1.5.3 Matbench and Magpie: the controlled benchmark contrast

Matbench entered after the raw-source experiments.
Its role was not to replace JARVIS or the Materials Project.
It provided a curated,
standardised benchmark environment in which to test whether the
central distinctions survived:

- community-standard target preparation;
- strong predictive performance;
- a neural learner;
- a high-dimensional correlated composition descriptor space;
- complete retraining after descriptor removal.

This made Matbench a controlled contrast rather than the origin of the
programme.
Raw Materials Project data exposed where the scientific failures
lived.
JARVIS supplied the first coherent cross-property and replay anchor.
Matbench tested whether the assurance question remained relevant after
the task had been curated into conventional benchmark form.
Table 1.2
summarises these non-substitutable roles.

Table 1.2: Scientific role of each data environment in the monograph.

Environment	Programme role	Principal contribution	Principal boundary
JARVIS-DFT	First complete four-target audit and replay anchor.	Demonstrates bounded carrier signal, target dependence, scale behaviour, model and split dependence, perturbation response and numerical replay.	Calculated source objects; no external experimental validation; cross-source transfer not established by this stage alone.
Materials Project	Cross-source extension and adverse-result environment.	Shows bounded energetic/electronic recurrence, elastic source-admissibility failure, relative-only model success and mixed replay states.	Raw elastic interpretations remain held; only a small exact-structure overlap supports direct matched-material comparison.
Matbench + Magpie	Curated benchmark and neural-transfer contrast.	Shows that strong predictive performance does not eliminate importance-necessity divergence; tests viable transformed elasticity.	Primarily a curated Materials Project-derived formulation, not a fully independent third physical source.

The ordering is important.

The monograph does not move from an untidy dataset towards a cleaner

one and then forget the original problem.

It uses each environment to answer a different question.

1.6 Research questions

The monograph is organised around six primary research questions.

1. **Carrier signal**

   Do the selected material targets contain sufficient non-null

   predictive structure under leakage-controlled tabular routes to support

   a meaningful descriptor audit?

2. **Conditional explanation**

   How strongly do descriptor accounts depend on target,

   model family,

population scale,

split geometry

and declared input disturbance?

3. **Importance versus necessity**

   Does a descriptor ranked as important remain consequential when it is removed and the full modelling process is retrained?

4. **Model viability**

   Is positive separation from a paired null sufficient for engineering interpretation,

   or must the model also cross an absolute usefulness threshold?

5. **Source and replay authority**

   How should source admissibility,

   semantic identity,

   numerical equivalence

   and raw-byte identity be separated when a completed execution returns mixed reproducibility states?

6. **Engineering permission**

   What bounded action,

   if any,

   can the resulting evidence support before a higher-fidelity calculation,

   physical test

   or human decision is required?

These questions are deliberately narrower than a claim of universal invariance.

The study tests persistence within declared transformation families.

It does not attempt to exhaust every possible representation,

learner,

dataset

or material domain.

1.7 Contributions

The monograph makes eight principal contributions.

1. **A materials-assurance framing.**
   The work positions model explanation as an engineering evidence problem rather than treating explanation as a decorative layer added after prediction.
2. **A formal distinction among evidence objects.**
   Predictive importance,
   transformation stability,
   retrained-removal consequence,
   model viability,
   source admissibility
   and replay equivalence are kept separate.
3. **Bounded descriptor authority.**
   The study defines authority as conditional permission for a declared claim within an evidenced domain,
   rather than as a universal property of a feature.
4. **A multi-axis descriptor-response object.**
   Descriptor behaviour is recorded across model,
   seed,
   scale,
   population,
   noise,
   dropout,
   shuffle,
   removal
   and source.
5. **A two-door model-viability rule.**
   A model must cross both an absolute predictive threshold and a paired null-separation threshold before bounded interpretation is permitted.
6. **A cross-source empirical programme.**

The method is exercised across JARVIS-DFT,
the Materials Project
and Materials Project-derived Matbench tasks,
while retaining the distinction between source recurrence and
matched-material transfer.

7. **Replay and recoverability as scientific properties.**
   Long-running executions are checkpointed at atomic scientific-unit
   boundaries,
   later executed as separately identified replays
   and judged separately for semantic,
   numerical
   and byte equivalence.
8. **An explicit claim ladder.**
   Observed,
   repeated,
   bounded,
   cross-source,
   mechanistic,
   causal
   and universal claims are separated so that no result is promoted
   beyond its evidence state.

1.8 Scope and claim boundary

The monograph supports a bounded materials-assurance method.
It reports local computational evidence that:

- formation energy,
  band gap,
  bulk modulus
  and shear modulus contain meaningful non-null signal within the
  declared JARVIS routes;
- descriptor behaviour changes with target,
  model,

scale,

split

and disturbance;

- formation-energy and band-gap behaviour shows bounded recurrence across JARVIS and Materials Project environments;
- raw Materials Project elasticity requires an explicit source-admissibility protocol before physical interpretation;
- positive null separation does not guarantee absolute model viability;
- strong neural prediction does not guarantee that a highly ranked descriptor is necessary after retraining;
- scientific numerical replay,

  semantic identity

  and raw-byte identity can return different states;
- long-running scientific programmes can resume from verified unit boundaries without silently accepting incomplete work.

The monograph does not establish:

- a universal hierarchy of material descriptors;
- physical causality from model attribution;
- experimental validity of the calculated records;
- material-by-material transfer between JARVIS and Materials Project;
- a production-qualified material-selection system;
- fabrication,

  certification

  or regulatory authority;
- a universal theory of system stability,

  drift,

  collapse

  or recovery;
- autonomous scientific or engineering decision authority.

Claim boundary

The strongest defensible contribution is not that Geatomica discovers physical truth automatically.
It is that the framework measures where a materials-model interpretation remains coherent,
where it depends on the route that produced it,
where it becomes replaceable,
and where the evidence requires the engineer to stop.

1.9 Structure of the monograph

The final publication arc is science first: Chapters 1–3 define the research object and experimental logic; Chapters 4–8 report JARVIS, Materials Project, Matbench and the spatial persistence/DOE/confirmation lineage; Chapter 9 separates the three publication questions; Chapters 10–11 describe methodological change and engineering decisions; Chapters 12–13 treat runtime, recovery, scientific memory and authority; and Chapters 14–15 set novelty, limitations and falsification routes. Appendices bind the experiment, claim and replay registers.

Chapter 2

Descriptor Authority and the Geatomica Research Object

Conceptual notation. Equations in this chapter organise evidence classes and successor hypotheses. They are not fitted empirical laws, calibrated universal scores, or validated physical equations unless a later empirical section explicitly reports the corresponding executable measurement.

2.1 The physical question beneath the modelling problem

The original Geatomica question was not whether machine learning could

predict a material property.

That had already been demonstrated extensively across composition,

structure,

electronic

and mechanical targets.

The more specific question was whether the dimensionality of those

problems could be reduced without reducing the resulting model to a

black-box shortcut.

A material response is produced through several connected physical

layers.

Composition constrains bonding possibilities.

Bonding and electronic state shape the available energetic landscape.

Energetic organisation constrains stable and metastable structures.

Structure and geometry determine packing,

coordination

and directional response.

Those responses eventually determine how the material behaves under

load,

temperature,

environment

and time.

This suggests a possible hierarchy:

$$\begin{aligned} \text{constitution and composition} &\rightarrow \text{bonding and electronic state} \\ &\rightarrow \text{energetic organisation} \\ &\rightarrow \text{structure and geometry} \\ &\rightarrow \text{packing and coordination} \\ &\rightarrow \text{material response} \\ &\rightarrow \text{damage and failure.} \end{aligned} \tag{2.1}$$

Equation 2.1 is not presented as a universal

one-way causal law.

It is a physically organised research hypothesis.

Its purpose is to structure experiments around questions such as:

- Do descriptors nearer to a target in the proposed chain remain more stable under modelling disturbance?
- Do electronic or energetic descriptors provide useful information about later structural or mechanical response?
- Are apparent upstream descriptors genuinely persistent, or do they only perform well inside one material family, database or representation?
- Can a smaller descriptor spine preserve predictive and diagnostic performance inside a declared domain?
- Where must a reduced model escalate to a richer structural, atomistic or experimental representation?

A negative result would still be scientifically useful.

If no stable hierarchy survives model,

population

or source changes,

then the appropriate conclusion is that descriptor order is local and

conditional.

Geatomica is therefore not committed to finding a universal spine.

It is committed to making the search falsifiable.

Physical-role chain boundary

The physical-role chain organises the experiment before terminal analysis.
It does not permit the model's final ranking to be narrated retrospectively as a causal chain.
A relationship becomes physically interesting only after surviving the relevant modelling tests and being connected independently to materials physics.

2.2 Why one fitted importance ranking is insufficient

Feature-importance methods answer useful questions.

Permutation importance measures how prediction changes when a feature

is disturbed within an already fitted model.

Tree-based impurity measures describe how a model used variables while

constructing its internal partitions.

Linear coefficients describe one fitted relation after accounting for

the chosen scaling,

regularisation

and correlated feature set.

Gradient-based attribution and SHAP-style methods distribute model

output across inputs under their own assumptions and reference states.

None of these quantities is meaningless.

The problem occurs when a conditional model description becomes a

physical claim without additional testing.

Suppose two descriptors,

x_1 and x_2,

contain strongly overlapping information.

A fitted model may route most of its prediction through x_1.

Perturbing x_1 after fitting can therefore cause a large loss.

The fitted model correctly reports that it relies on x_1.

However,

if x_1 is removed before training,

the model may learn to reconstruct almost the same prediction through

x_2.

The first experiment measures fitted reliance.

The second measures retrained-removal consequence within the available

representation.

Both results can be true.

They answer different questions.

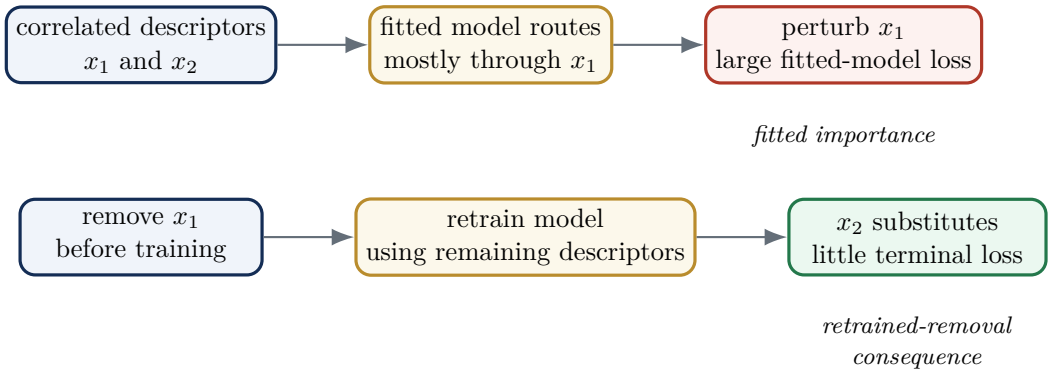


Figure 2.1: Fitted importance and retrained-removal consequence answer different questions. A descriptor can be central to the information route used by one trained model while remaining replaceable when the model is trained again without it.

Figure 2.1 illustrates the central

distinction.

A highly ranked descriptor may be:

- a physically meaningful variable;
- a strong fitted-model route;
- a stable correlate across several runs;
- yet still replaceable by other descriptors after retraining.

Conversely,

a quieter descriptor may appear less important in the fitted model but

prove difficult to replace when removed.

The brightest part of the explanation is not automatically the most

load-bearing part of the representation.

Governing rule

Importance describes a model's present information route.
Retrained removal tests what the modelling process can reconstruct when that route is removed.
Authority cannot be inferred from either quantity alone.

2.3 Descriptor authority as bounded engineering permission

Geatomica defines descriptor authority operationally.

For descriptor j,

target t

and evidenced domain $\mathcal{D}$,

authority is not a claim that the descriptor is universally true.

It is permission to use a specified relationship for a specified

purpose within a bounded domain.

A general representation is:

$$\mathcal{A}_{j,t,\mathcal{D}} = \Phi\left(P, N, T, M, S, D, R, C\right), \tag{2.2}$$

where:

- P denotes predictive usefulness;
- N denotes separation from negative controls;
- T denotes transformation stability;
- M denotes recurrence across model families;
- S denotes recurrence across scale and population split;
- D denotes source and domain admissibility;
- R denotes replay and reproducibility evidence;
- C denotes the claim or decision being requested.

Equation 2.2 is intentionally not reduced to a

single universal scalar.

Different engineering decisions place different demands on evidence.

A descriptor used to organise exploratory analysis does not require

the same evidence as one used to remove physical tests from a

qualification programme.

A relationship used to nominate a higher-fidelity calculation does

not require the same authority as one used to reject fabrication.

The claim is therefore part of the authority object.

Formally,

a permission can be written as:

$$\mathcal{P} = (q, \mathcal{D}, \mathcal{T}, \mathcal{E}, a)\,, \tag{2.3}$$

where:

- q is the claim;
- $\mathcal{D}$ is the evidenced domain;
- $\mathcal{T}$ is the declared transformation family;
- $\mathcal{E}$ is the supporting evidence bundle;
- a is the permitted action.

Examples of actions include:

- retain for exploratory interpretation;
- nominate for a higher-fidelity calculation;
- use within a bounded screening model;
- request additional source resolution;
- hold interpretation;
- refuse downstream promotion.

Engineering consequence

Descriptor authority is meaningful only when attached to a decision.
Without a declared action,
"authority" becomes an abstract label.
With a declared action,
it becomes testable:
the evidence either supports that permission or it does not.

2.4 The descriptor-response barcode

A single ranking is too compressed to represent the evidence required

for authority.

Geatomica instead records a descriptor-response object.

For descriptor j,

target t

and material region z:

$$\mathbf{b}_{j,t,z} = [I, \Delta_{\text{noise}}, \Delta_{\text{dropout}}, \Delta_{\text{shuffle}}, \Delta_{\text{removal}}, L_{\text{scale}}, L_{\text{split}}, R_{\text{model}}, \Delta_{\text{source}}] . \tag{2.4}$$

The components are:

I	Baseline fitted importance under a declared model, population and explanation method.
Δ_{noise}	Change under controlled numerical disturbance.
Δ_{dropout}	Change when a declared fraction of eligible inputs is removed or made missing.
Δ_{shuffle}	Loss after disrupting the association between one descriptor and the other rows while retaining its marginal distribution.
Δ_{removal}	Performance consequence after the descriptor is removed and the model is trained again.
L_{scale}	Change across deterministic nested population sizes.
L_{split}	Loss under increasingly difficult population-separation rules.
R_{model}	Recurrence across model families and seeds.
Δ_{source}	Change across independently governed data environments.

The barcode does not assume that every component should move in the same direction.

A descriptor can show:

- high baseline importance;
- low noise sensitivity;
- large shuffle loss;
- small retrained-removal loss;
- strong recurrence across tree models;
- weak recurrence in a linear model;
- good within-source persistence;
- poor cross-source persistence.

That pattern is more informative than a terminal average.

It describes the shape of the evidence.

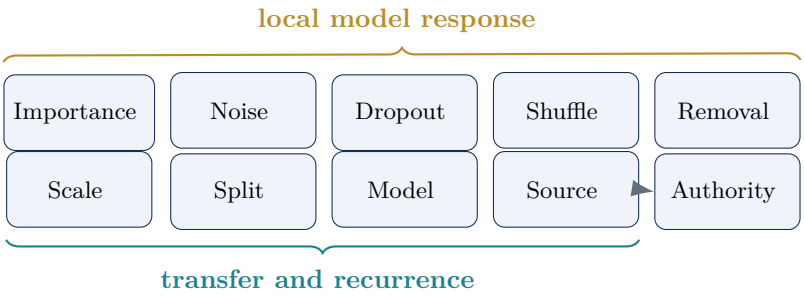


Figure 2.2: Conceptual descriptor-response barcode. The barcode keeps fitted response, disturbance response, retrained consequence and transfer evidence separate before a bounded authority decision is made.

Descriptors can be compared in the response space using:

$$d(j,k) = \sqrt{(\mathbf{b}_j - \mathbf{b}_k)^{\mathrm{T}}\, \mathbf{W}\, (\mathbf{b}_j - \mathbf{b}_k)}, \tag{2.5}$$

where $\mathbf{W}$ is a declared weighting matrix.

The weighting must be frozen before terminal classification or

estimated through a separately governed procedure.

Unweighted analysis and sensitivity to $\mathbf{W}$ should be retained.

Otherwise,

the apparent geometry can be manufactured by convenient weighting.

2.5 Bounded invariance rather than universal invariance

The term *invariance* can easily become stronger than finite

experiments permit.

No finite audit over a small family of transformations can establish

that a relation survives every possible admissible change.

The defensible object is bounded invariance under a declared

transformation family.

Let $\mathcal{T}^*$ be the finite set of transformations used by the

experiment.

For a quantity q:

$$I_{\mathcal{T}^*}(q) = \Pr\left[q\text{remains within its declared validity bounds under}\tau \in \mathcal{T}^*\right]. \tag{2.6}$$

The empirical programme estimates this bounded object.

It does not establish:

$$q\text{is invariant under all conceivable transformations.} \tag{2.7}$$

This produces an evidence ladder:

1. **Observed**

   The result appeared in one fitted configuration.

2. **Repeated**

   The result recurred across seeds or executions.

3. **Boundedly persistent**

   The result survived a declared transformation family.

4. **Cross-source persistent**

   The result recurred across provenance-separated source environments.

5. **Mechanistically supported**

   The relation was connected independently to physical intervention,

   simulation,

   measurement

   or established domain theory.

6. **Causal**

   The relation survived an appropriate causal or interventional test.

7. **Universal**

   The relation is claimed to hold beyond the bounded evidence domain.

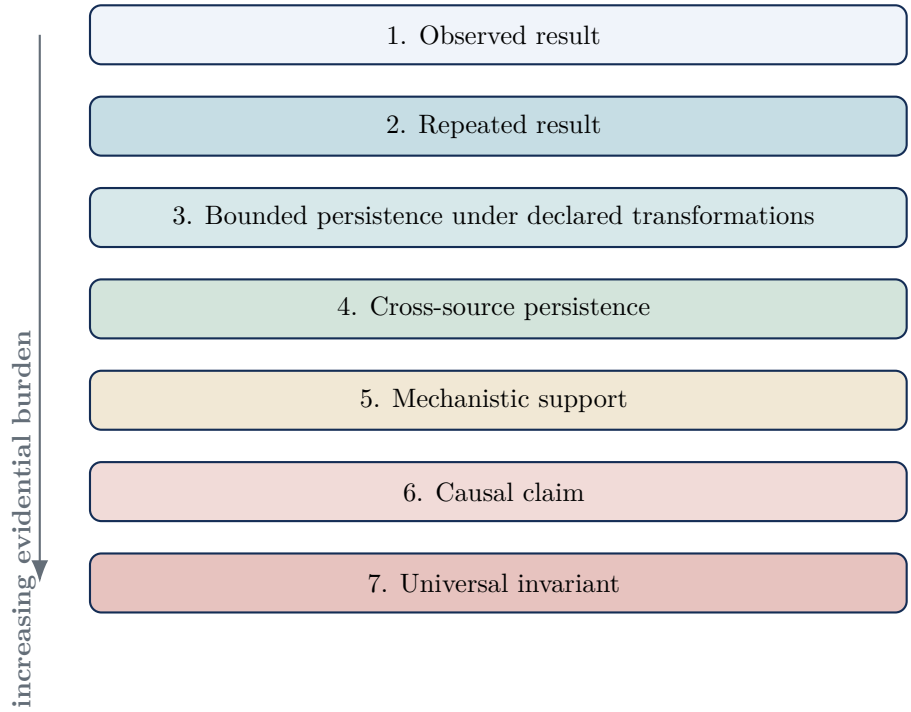


Figure 2.3: Evidence ladder used to prevent bounded computational persistence from being promoted directly into causal or universal language. The present programme reaches different levels for different results.

Governing rule

Every claim inherits the ceiling of its weakest required evidence object.
A later interpretation cannot repair an earlier failed source, viability
or replay gate.

2.6 The two named operational components

2.6.1 Governed Invariance Logic

GIL is the audit logic applied to a declared scientific object.

Its function is to:

- freeze the source,

  target,

  population

  and descriptor conditions;

- declare the transformation family before terminal interpretation;

- apply controlled changes to model,
  scale,
  split,
  input quality
  and descriptor availability;
- compare valid models with paired negative controls;
- record persistence,
  deformation,
  substitution
  and collapse;
- produce bounded evidence objects rather than one universal score.

GIL is not an autonomous truth engine.

It measures response under declared conditions.

The logic can be represented as:

$$(Q_{\text{evidence}}, U, \Pi_{\text{audit}}) \xrightarrow{x_{\text{GIL}}} G_{\text{authority}}, \tag{2.8}$$

where:

- Q_{evidence} is the high-dimensional evidence state;
- U defines source,
  population,
  representation
  and experimental boundaries;
- Π_{audit} is the declared pressure or perturbation vector;
- x_{GIL} is the controlled transformation and measurement
  route;
- $G_{\text{authority}}$ is the lower-dimensional evidence geometry
  used for bounded adjudication.

This is an operational research design,

not a claim that the geometry is a new physical ontology.

2.6.2 Bounded candidate progression

Candidate progression was organised using bounded multi-objective selection over predictive error, stability, complexity, compute and declared risk.

The purpose was not to let one best-score criterion silently define scientific progress. Candidate options remained subject to the same frozen admission, viability, perturbation, replay and claim-boundary checks described elsewhere in this monograph.

This public evidence release reports the scientific questions, measured outcomes and bounded decision consequences needed to inspect the claims. The reusable progression implementation, advancement policy and production orchestration are outside the public evidence release. No candidate-selection mechanism was permitted to grant engineering authority.

Why use bounded multi-objective progression?

The experiment contains competing scientific and operational considerations. A candidate that improves one score may become less stable, more expensive, more complex or less capable of exposing a known failure mode. Keeping those considerations visible prevents predictive improvement from impersonating complete scientific progress.

2.7 Supporting functions without architecture takeover

Four further functions support the experiment.

They are described by what they do rather than introduced as additional

frameworks.

1. **Visual evidence representation**

   The response surface,

   Pareto trade-offs,

   descriptor zones,

   source holds

   and replay states must be visible to a human reviewer.

   The visual layer does not change evidence.

   It makes the current evidence geometry inspectable.

2. **State memory and provenance**

   The programme retains protocol identity,

   source identity,

   unit receipts,

   checkpoints,

   hashes,

   terminal manifests,

   accepted evidence

   and rejected routes.

Narrative summaries are not treated as the authoritative source.

3. **Operational governance**

   The operating layer controls which module may execute,

   what evidence may be promoted,

   which state applies

   and what claim ceiling remains in force.

4. **Recovery control**

   Long-running work is divided into complete scientific units.

   Only units satisfying the expected-output contract are committed.

   Interrupted work resumes from the last verified boundary.

These functions become important later in the empirical programme.

They are not the scientific protagonist.

The scientific protagonist remains the materials relationship being tested.

2.8 Chapter conclusion

Descriptor authority emerged because the experiments required a stronger concept than fitted importance or repeated stability alone.

The framework preserves several distinctions:

- importance versus necessity;
- prediction versus viability;
- repetition versus bounded persistence;
- cross-source recurrence versus matched-material transfer;
- numerical replay versus byte identity;
- evidence generation versus engineering permission.

The next chapter converts these distinctions into a frozen experimental programme.

The central question is no longer only whether a descriptor looks important.

It is whether its apparent role survives enough independent pressure to support a bounded materials-engineering claim.

Chapter 3

Experimental Programme and Decision Logic

3.1 Programme architecture

The empirical programme was designed to expose disagreements that a single benchmark score would hide.

The principal axes were:

- target property;
- source environment;
- descriptor lane;
- population scale;
- split geometry;
- model family;
- random seed;
- negative control;
- input perturbation;
- descriptor removal;
- fresh versus replay execution.

The programme was not designed as an exhaustive search over every possible material representation.

It was designed as a controlled first measurement of whether descriptor interpretations persist across several scientifically meaningful changes.

The experimental progression was:

$$\text{JARVIS} \rightarrow \text{Materials Project} \rightarrow \text{Matbench and Magpie.} \tag{3.1}$$

Each stage inherited lessons from the earlier one,

but historical runs were not silently rewritten.

Corrections were introduced through named successor protocols.

3.2 Target selection

Four targets were selected to sample different positions in the

proposed materials chain.

Table 3.1: Targets used in the four-property programme.

Target	Unit	Physical role	Reason for inclusion
Formation energy per atom	eV/atom	Energetic organisation and relative thermodynamic state.	Tests whether composition, structure and related property information reconstruct energetic organisation.
Band gap	eV	Electronic state and separation between occupied and unoccupied electronic levels.	Provides an electronic target with known dependence on composition, structure, method and material class.
Bulk modulus	GPa	Resistance to uniform volumetric deformation.	Tests recovery of a scalar mechanical response strongly connected to bonding, packing and equation-of-state curvature.
Shear modulus	GPa	Resistance to shape change at approximately constant volume.	Provides a more directionally sensitive mechanical target expected to be less accessible through coarse tabular descriptors.

Formation energy and band gap used the largest available coherent

populations.

The elastic targets were limited to records exposing the relevant

source quantities.

No target was assumed to be automatically admissible because the field

was present.

3.3 Descriptor lanes

Two principal lanes were used.

3.3.1 Common descriptor lane

The common lane provided a shared comparison substrate across all four targets.

It included a conservative set of descriptors such as:

- density;
- volume per atom;
- site count;
- total magnetisation;
- lattice lengths;
- lattice angles;
- crystal system.

The purpose of the common lane was not to maximise predictive performance.

It allowed the same broad representation to be compared across energetic, electronic and elastic targets.

3.3.2 Target-specific non-leaking lane

The target-specific lane added related properties only where a frozen exclusion register permitted them.

Examples included energetic or electronic information when not directly equivalent to the target.

The lane excluded:

- the target itself;
- direct target siblings;
- post-target labels;
- derived quantities that reconstructed the target trivially;
- identifiers or source artefacts;
- features listed in the leakage register.

The label *target-specific* does not mean causal.

It means the descriptor set contains additional information relevant

to that target under a frozen non-leakage contract.

Table 3.2: Purpose and boundary of the two descriptor lanes.

Lane	Purpose	Boundary
Common	Compare targets under a shared, conservative tabular representation.	May omit useful target-specific information; does not represent the strongest possible predictor.
Target-specific	Test whether admissible related properties improve prediction and change descriptor response.	Measures information transfer, not causality; every target requires a separate exclusion register.

3.4 Population scales

Deterministic nested cohorts were constructed at:

$$n \in \{2{,}000, 5{,}000, 10{,}000, n_{\text{full}}\}\,. \tag{3.2}$$

The smaller cohorts were nested within the larger populations.

This allows scale response to be interpreted without replacing the

underlying population arbitrarily at each size.

The scale experiment asks:

- Does predictive signal improve as more records become available?
- Does descriptor ordering stabilise with scale?
- Do some descriptors appear only at small or large population sizes?
- Does performance convergence coincide with explanation convergence?

The final question is especially important.

A model can improve steadily with more data while the apparent

descriptor account remains conditional on model or split.

3.5 Population splits

Three split views were used.

3.5.1 Row-random split

Rows were divided randomly under a fixed seed and test fraction.

This is the least demanding view.

It primarily tests interpolation across records drawn from a similar

overall population.

3.5.2 Reduced-composition group split

Records sharing the same reduced composition were assigned to the same group.

This prevents identical reduced formulae from appearing on both sides of the split.

It asks whether the model transfers to unseen compositions rather than only unseen records.

3.5.3 Chemical-system group split

Records sharing the same element set were assigned to the same group.

This is a harder population shift.

It asks whether the learned relation persists when complete element combinations are withheld.

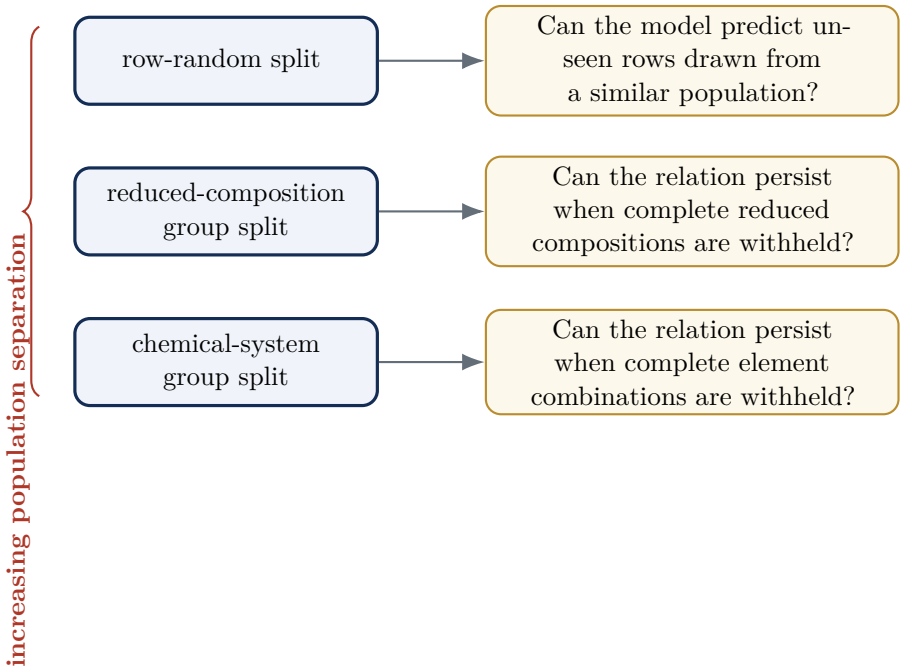


Figure 3.1: The three split views represent different scientific claims. Performance under row-random separation cannot be treated as evidence of unseen-composition or unseen-chemical-system transfer.

Governing rule

The split is part of the claim.
Prediction of unseen rows,
unseen compositions
and unseen element systems are not interchangeable forms of generalisation.

3.6 Model families

The first two source programmes used four model families:

1. **Ridge regression**

   A regularised linear baseline.

2. **Random Forest**

   A nonlinear bagged tree ensemble.

3. **Extra Trees**

   A highly randomised nonlinear tree ensemble.

4. **HistGradientBoosting**

   A sequential nonlinear boosting model.

The model set was selected to compare:

- linear and nonlinear information routing;
- bagged and boosted ensembles;
- different sensitivity to feature scaling,

  interactions

  and threshold structure;
- different explanation behaviours under permutation,

  shuffle

  and removal.

The aim was not to identify one universally best model.

It was to determine whether conclusions survived materially different learners.

A later multilayer perceptron added a distributed nonlinear learner in the Matbench/Magpie stage.

3.7 Preprocessing and leakage control

All preprocessing was fitted using the training population only.

Numeric descriptors were:

- median-imputed from the training set;
- standardised where required by the model family;
- kept in a floating-point working frame during dropout so that missing values could be represented without mutating integer source columns.

Categorical descriptors such as crystal system were:

- most-frequent imputed using the training set;
- one-hot encoded;
- configured to ignore categories unseen during training.

The leakage scan included:

- explicit forbidden columns;
- identifier-like names;
- post-target fields;
- direct target derivatives;
- suspiciously perfect target correlations;
- source-specific leakage routes.

All transformations used during fitting were contained inside the training pipeline.

3.8 Negative controls

Each full-scale valid model cell was paired with target-permutation nulls.

The training target was shuffled while retaining:

- the source cohort;
- the split;
- the descriptor matrix;
- the preprocessing route;
- the model family;
- the seed context.

This produced a paired null score:

$$R^2_{\text{null}}. \tag{3.3}$$

The valid-minus-null separation was:

$$S = R^2_{\text{valid}} - R^2_{\text{null}}. \tag{3.4}$$

A positive S indicates that the valid target carries more learnable structure than the paired shuffled target under that cell.

It does not establish that the valid model is useful.

3.9 The two-door model-viability rule

The Materials Project experiment exposed a defect in the original aggregation logic.

A model could satisfy:

$$S > 0 \tag{3.5}$$

while still satisfying:

$$R^2 < 0. \tag{3.6}$$

The model had beaten the deliberately weakened comparator while remaining worse than predicting the mean.

The corrected rule requires:

$$\text{viable support} \iff \left(R^2 > \tau_{\text{absolute}}\right) \wedge \left(S > \tau_{\text{separation}}\right). \tag{3.7}$$

For the baseline exposed in this monograph:

$$\tau_{\text{absolute}} = 0, \qquad \tau_{\text{separation}} = 0. \tag{3.8}$$

Table 3.3: Two-door model-viability decisions.

State	$R^2 > 0$	$S > 0$	Bounded decision
Absolute and relative pass	Pass	Pass	Candidate may be nominated for bounded interpretation.
Relative-only pass	Fail	Pass	**[HOLD]**. Signal may exist, but the model is not absolutely viable.
Absolute-only pass	Pass	Fail	**[HOLD]**. The model has not separated from the paired null under the declared control.
Both fail	Fail	Fail	**[REFUSE]** predictive interpretation.

Engineering consequence

A broken thermometer may outperform a random-number generator while remaining a broken thermometer.
Relative improvement is evidence.
It is not viability.

3.10 Perturbation routes

The declared perturbation family included:

3.10.1 Numerical noise

Gaussian noise was applied to eligible numeric descriptors at a

declared fraction of each descriptor's observed standard deviation.

The primary JARVIS route used 2% noise.

This tests local numerical sensitivity.

It is not a model of physical ageing,

oxidation,

fatigue

or measurement degradation.

3.10.2 Descriptor dropout

A declared fraction of eligible numeric cells was replaced with

missing values.

The primary full-scale route used 10% independent cell dropout.

This tests tolerance to incomplete descriptor availability.

3.10.3 Within-feature shuffle

One descriptor was shuffled across rows.

Its marginal distribution was retained,

but its association with the target and other row-level information

was disrupted.

3.10.4 Permutation importance

Prediction loss was measured after feature permutation in the fitted

model.

3.10.5 Complete retrained removal

A descriptor or descriptor group was removed before fitting.

The preprocessing and model were reconstructed using the remaining

representation.

This measures substitutability within the available feature space.

3.11 Metrics and aggregation

The executable routes used scikit-learn's `r2_score(y_true,y_pred)`:

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y}_{\text{eval}})^2},$$

where $\bar{y}_{\text{eval}}$ is the mean of the evaluated held-out targets. Earlier training-mean wording was a manuscript error. Positive R^2 is the minimal demonstrated absolute boundary against this constant reference in the present failure-analysis route, not a universal engineering threshold. MAE, RMSE, null gaps, fold recurrence and declared perturbation effects remain separate evidence objects.

Aggregation rules were frozen before result inspection. Split views are reported separately where the protocol requires separate adjudication.

3.12 Replay contracts

Replay was adjudicated through separate contracts.

1. **Plan identity**

   The same ordered scientific-unit plan must be executed.

2. **Source and configuration identity**

   Dataset,

   code,

   configuration,

   dependency

   and preprocessing identities must match the frozen contract.

3. **Schema identity**

   Expected files,

rows,

columns

and identifiers must be present.

4. **Semantic identity**

   Corresponding values must refer to the same scientific objects.

5. **Numerical equivalence**

   Compared numeric values must remain within the frozen tolerance.

6. **Raw-byte identity**

   Corresponding files must be byte-identical.

The principal numerical tolerance was:

$$|x_{\text{fresh}} - x_{\text{replay}}| \leq 10^{-10}. \tag{3.9}$$

Byte identity was judged separately.

A numerically equivalent result was not described as byte-identical

when the byte contract failed.

3.13 Atomic scientific units and recovery

The execution controller used:

- a canonical ordered unit plan;
- single-writer locking;
- append-only journals;
- atomic unit commits;
- expected-output contracts;
- hash verification;
- incomplete-unit rejection;
- resume under the same execution identity;
- terminal full-disk verification.

Scientific unit: one complete, independently committable part of the declared experiment. It may contain one estimator fit, several fits, a control set, or a solver/scenario calculation. A complete scientific unit was the smallest accepted restart boundary.

An interrupted staging unit was not promoted merely because some files

existed.

This distinction protects the scientific record from partial

completion being mistaken for accepted evidence.

3.14 Chapter conclusion

The experimental design treats source,

model,

split,

scale,

perturbation

and replay as parts of the scientific object.

It does not ask only whether prediction survives.

It asks which parts of the explanatory account survive,

which parts are reconstructed through alternatives,

and which claims become inadmissible.

The first full application of this design was the JARVIS-DFT

programme.

Chapter 4

JARVIS-DFT: The First Complete Audit

Question. Can compact descriptors carry useful signal across targets, models, population splits, scale, and perturbation?
Comparison. Four targets were tested across common and target-specific descriptors, four classical models, three seeds, three split strategies, four scales, perturbations, removals, and paired target-permutation nulls.
Result. Useful but target-, model-, and population-dependent signal was observed; the 48-unit replay was numerically equivalent across 28,248 cells (maximum difference approximately $1.07 imes 10^{-14}$).
Decision. Retain JARVIS as the coherent baseline and as evidence that null separation and useful prediction can coexist.
Boundary. Local measured and same-environment replayed evidence; not external validation or physical mechanism proof.

4.1 Role of the JARVIS stage

JARVIS-DFT supplied the first complete execution of the four-target

programme.

Its scientific role was to establish:

- whether sufficient predictive carrier signal existed across energetic,

  electronic

  and elastic targets;
- whether target-specific information improved performance without

  leakage;
- whether model,

  scale

  and population split changed the apparent descriptor account;
- whether controlled noise and dropout exposed different forms of

  fragility;
- whether the complete scientific programme could be independently

replayed under a frozen contract.

The JARVIS stage was not designed to prove cross-source transfer.

It established the first bounded evidence surface against which the

Materials Project and Matbench stages could later be compared.

4.2 Source census and cohorts

The processed JARVIS-DFT `dft_3d` source census contained:

$$75{,}993 \tag{4.1}$$

calculated records.

Formation-energy and band-gap maximum-coverage cohorts retained up to

the full source census.

The four-property common and elastic cohort contained:

$$23{,}824 \tag{4.2}$$

records with the required target availability.

The source snapshot retained negative elastic values rather than

silently removing them.

The processed population included:

- 277 negative bulk-modulus values;
- 1,951 negative shear-modulus values.

These values were retained in the original execution.

A separate sensitivity protocol was required before any later decision

to exclude or classify them.

The JARVIS records are calculated database objects.

They are not independently measured specimens.

Table 4.1: JARVIS source and execution summary.

Object	Observed value	Meaning
Processed source census	75,993 records	Full local JARVIS-DFT census used by the source freeze.
Four-property common cohort	23,824 records	Population exposing formation energy, band gap, bulk modulus and shear modulus under the frozen membership rules.
Fresh scientific units	48/48	Complete and terminally sealed.
Replay scientific units	48/48	Complete and independently sealed.
Model-result rows	1,728	Baseline, scale, model, seed and split results.
Descriptor-response rows	1,860	Importance, disturbance and removal responses.
Fresh wall time	9.93 h	Eight-worker workstation-scale execution.

4.3 Full-scale predictive carrier signal

Extra Trees produced the strongest full-scale mean performance in each

target/lane summary.

These values establish a usable carrier signal for the audit.

They do not establish state-of-the-art prediction.

Table 4.2: Full-scale JARVIS Extra Trees performance. Means and standard deviations are calculated over the declared seeds and split views.

Target	Lane	Mean R^2	SD	MAE	R^2–null
Formation energy	Common	0.703	0.021	0.387	0.822
Formation energy	Target-specific	0.759	0.016	0.349	0.876
Band gap	Common	0.593	0.044	0.418	0.704
Band gap	Target-specific	0.759	0.041	0.313	0.849
Bulk modulus	Common	0.759	0.017	20.095	0.883
Bulk modulus	Target-specific	0.780	0.017	18.850	0.890
Shear modulus	Common	0.607	0.040	15.102	0.730
Shear modulus	Target-specific	0.679	0.041	13.332	0.786

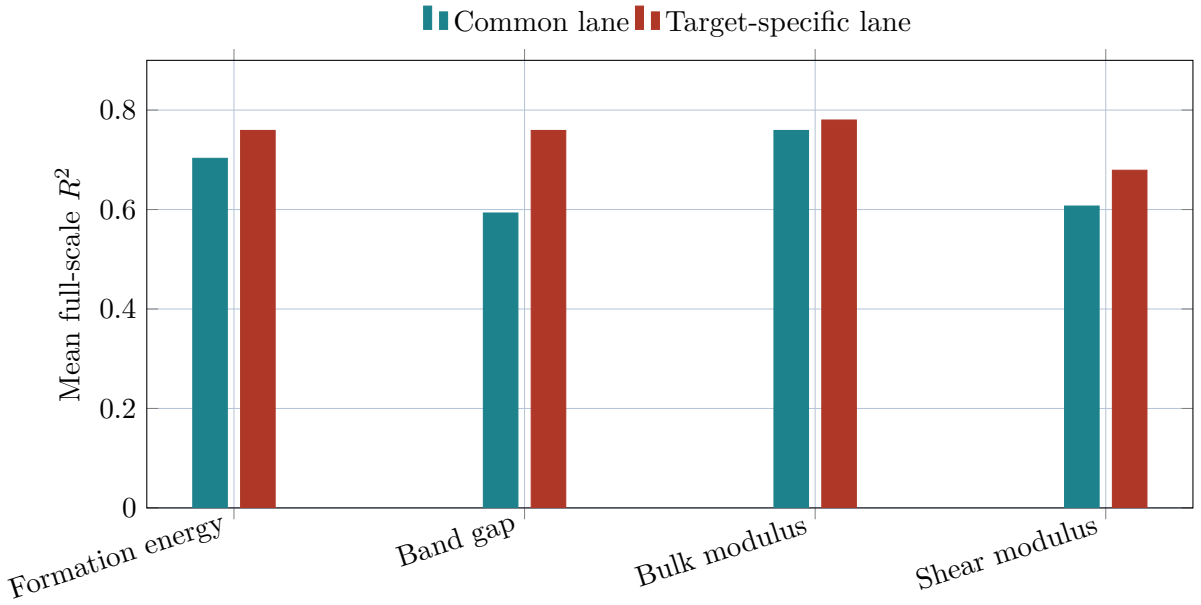


Figure 4.1: Full-scale JARVIS Extra Trees carrier signal by target and descriptor lane. Target-specific information improved all four targets, with the largest absolute gain for band gap.

Three principal patterns appear.

First,

bulk modulus was more accessible than shear modulus under the common

tabular route.

This is physically plausible.

Bulk response is a volumetric aggregate that can be strongly connected

to bonding strength,

density

and packing.

Shear response is more sensitive to directional bonding,

symmetry,

slip geometry

and anisotropy,

much of which is compressed or absent in coarse tabular descriptors.

This interpretation remains provisional.

The experiment supports target-dependent recoverability.

It does not prove the underlying mechanism.

Second,

target-specific information improved all four targets.

The largest gain was observed for band gap:

$$0.593 \rightarrow 0.759. \tag{4.3}$$

This indicates that the common lane omitted substantial

target-relevant information.

Third,

every full-scale valid cell with a paired null showed positive

valid-minus-null separation.

The JARVIS routes therefore contained meaningful non-null signal under

the tested conditions.

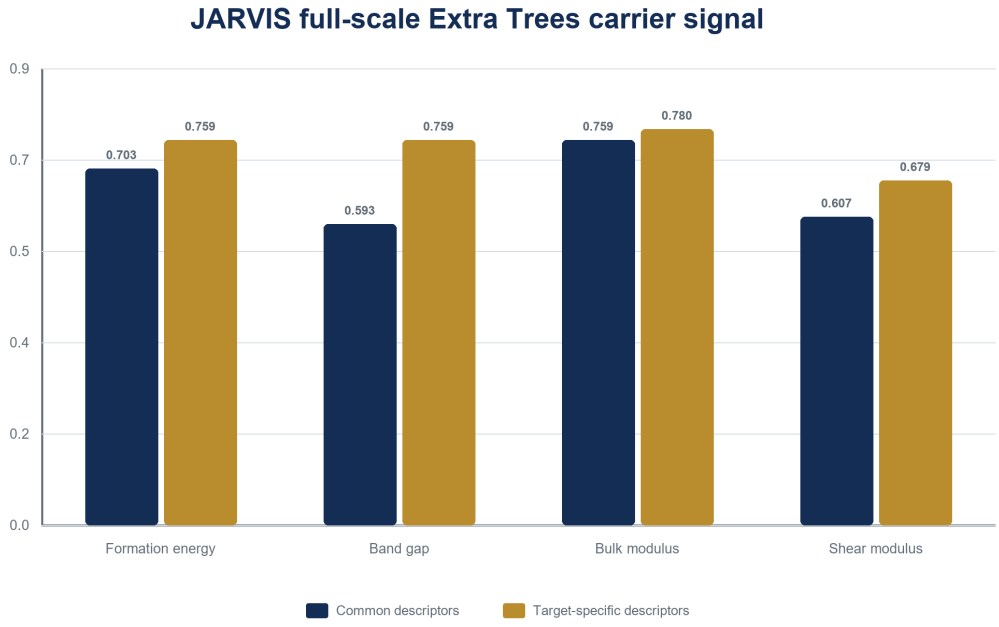


Figure 4.2: **HISTORICAL REPORT BOUND.** Full-scale Extra Trees carrier signal for the common and target-specific JARVIS descriptor lanes. The plotted values are reproduced from the claim ledger whose rows bind the accepted estate-audit report hash.

4.4 Scale convergence

Median R^2 increased from 2,000 rows to the full admissible

population for all eight target/lane aggregates.

Common-lane median values increased approximately as follows:

$$\text{formation energy:} \quad 0.498 \rightarrow 0.640, \tag{4.4}$$

$$\text{band gap:} \quad 0.355 \rightarrow 0.546, \tag{4.5}$$

$$\text{bulk modulus:} \quad 0.647 \rightarrow 0.742, \tag{4.6}$$

$$\text{shear modulus:} \quad 0.421 \rightarrow 0.574. \tag{4.7}$$

Larger populations improved prediction.

They did not eliminate model or split dependence.

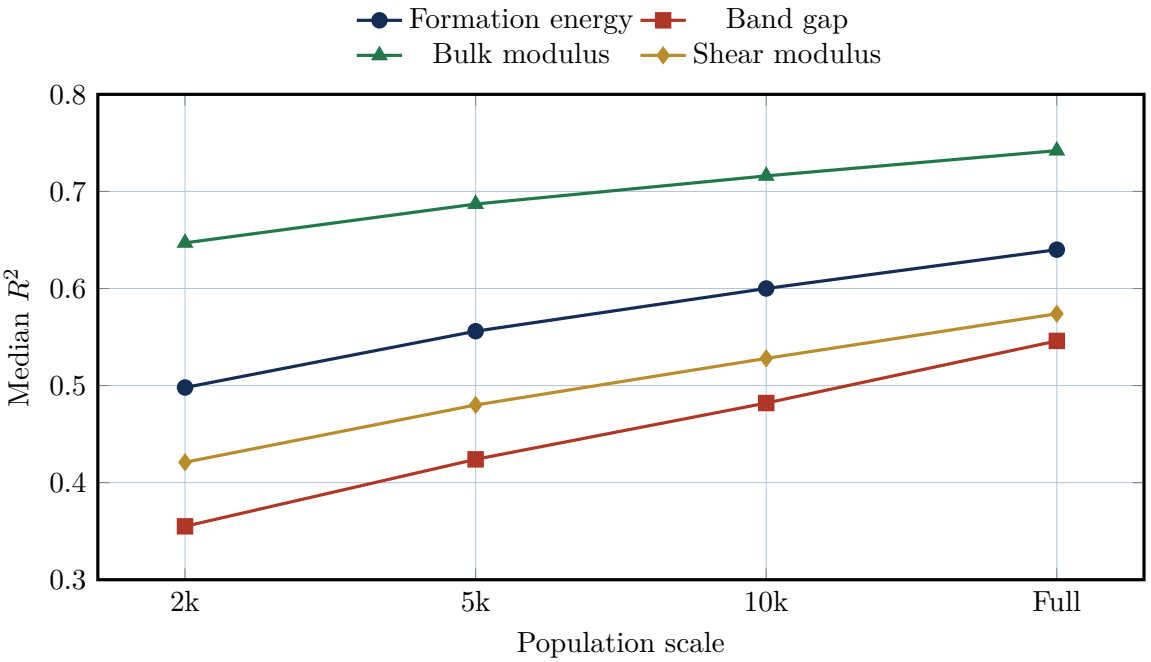


Figure 4.3: Common-lane scale response in JARVIS across 2k, 5k, 10k and full-scale cohorts ($N = 23{,}824$). All four targets show monotonic performance scaling as training population expands, with the highest relative sensitivity observed for band gap.

The scale result supports two conclusions.

First,

the signal was not confined to one small pilot cohort.

Second,

performance convergence did not settle the descriptor account.

The larger populations still produced material differences across

models and population splits.

4.5 Population-split dependence

Grouped splits generally reduced target-specific performance relative to row-random separation.

The hardest grouped view depended on the target.

Chemical-system separation was most costly for:

- formation energy;
- band gap.

Reduced-composition grouping was slightly more costly for:

- bulk modulus;
- shear modulus.

This prevents a universal ordering in which one split is always hardest.

The result is target-dependent.

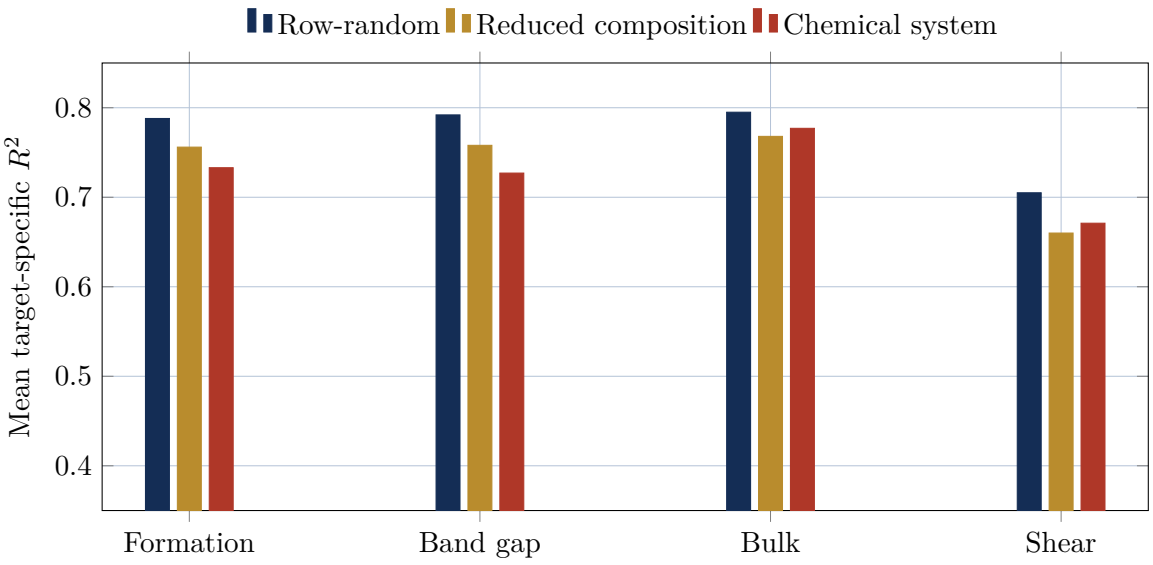


Figure 4.4: Target-specific split dependence in JARVIS across row-random, reduced-composition and chemical-system splits. Grouped splits reduce mean predictive performance relative to row-random splits for all four targets. Chemical-system separation is harder for formation energy and band gap; reduced-composition grouping is slightly harder for bulk and shear modulus.

The grouped-split result has a direct engineering meaning.

A model that performs well when chemically related records occur on both sides of the split should not automatically guide decisions about unseen chemical systems.

The population boundary must follow the intended use.

4.6 Model-family dependence

Ridge remained materially below the nonlinear ensembles.

In the common lane,

full-scale mean R^2 values were:

Table 4.3: Common-lane Ridge and Extra Trees comparison.

Target	Ridge	Extra Trees	Difference
Formation energy	0.244	0.703	0.459
Band gap	0.247	0.593	0.346
Bulk modulus	0.476	0.759	0.283
Shear modulus	0.256	0.607	0.351

The difference suggests that the recoverable information is

substantially nonlinear and interaction-dependent.

It does not mean that tree structure is the physical mechanism.

Tree ensembles can also exploit:

- correlated proxies;
- local population structure;
- threshold artefacts;
- redundancy;
- source-specific quirks.

Their strong performance strengthens the case for cross-model

descriptor auditing.

It does not grant the tree model's explanation privileged physical

authority.

Why retain a weak linear model?

The purpose of Ridge is not to compete for the highest score.
It provides a deliberately different information-routing regime.
When linear and nonlinear learners disagree,
that disagreement is evidence about the conditional nature of the
descriptor account.
Removing the weaker model would make the reported explanation appear
more coherent by construction.

4.7 Noise and dropout response

At full scale,

2% numerical noise produced Extra Trees mean R^2 losses of

approximately:

$$0.002\text{to}0.037. \tag{4.8}$$

Ten-percent descriptor dropout produced larger losses of:

$$0.066\text{to}0.106. \tag{4.9}$$

The two disturbances answer different questions.

Small numerical noise tests local smoothness around the observed

descriptor values.

Dropout tests dependence on descriptor availability.

A model can be robust to small numerical disturbance while depending

strongly on complete descriptor coverage.

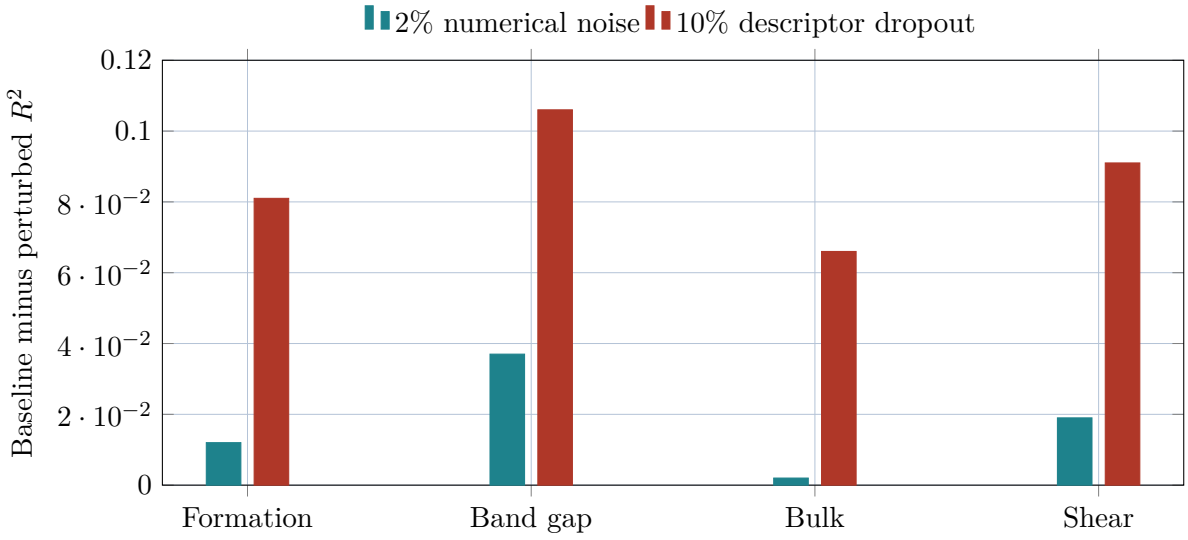


Figure 4.5: JARVIS Extra Trees perturbation response under 2% Gaussian noise and 10% descriptor dropout. Input noise produces moderate performance degradation ($\Delta R^2 \leq 0.037$), whereas descriptor dropout causes substantial accuracy loss ($\Delta R^2 \geq 0.066$) across all property targets.

Target-specific descriptors increased baseline signal.

In some routes,

they also increased dropout dependence.

This shows that added predictive information is not automatically added robustness.

4.8 Initial descriptor-response patterns

The leading common-lane responses differed by target.

The recorded patterns included:

- formation energy:
  site count,
  volume per atom
  and density;
- band gap:
  site count,
  density
  and magnetisation;
- bulk and shear modulus:
  volume per atom
  and density among the strongest responses;
- target-specific band gap:
  formation energy as a leading response.

These patterns are compatible with a conditional physical chain:

- composition and energetic organisation influence electronic state;
- packing proxies carry information about volumetric and shear response;
- different properties expose different accessible parts of the representation.

Compatibility is not causality.

The result identifies candidate relationships for stronger tests.

It does not establish a universal hierarchy.

Table 4.4: Bounded interpretation of initial JARVIS descriptor-response patterns.

Target	**Observed leading responses**	**Bounded interpretation**
Formation energy	Site count, volume per atom, density.	Energetic prediction uses composition and packing-related routes; necessity and causal direction remain unestablished.
Band gap	Site count, density, magnetisation; formation energy in the target-specific lane.	Electronic prediction benefits from energetic and structural context; the result is target- and model-conditioned.
Bulk modulus	Volume per atom and density.	Packing-related variables carry accessible information about volumetric stiffness.
Shear modulus	Volume per atom and density, with weaker total recovery.	Coarse packing proxies retain signal, but directional mechanical behaviour remains less accessible.

4.9 Fresh execution and predecessor-linked replay

The fresh execution completed:

$$48/48 \tag{4.10}$$

scientific units.

The predecessor-linked replay completed the same ordered plan under a

distinct execution identity.

The replay was interrupted during execution.

It resumed from verified controller-native checkpoints.

Only complete committed units were retained.

4.10 Comparator defect and correction

During replay adjudication,

a comparator defect was found.

A fallback identity key omitted descriptor-level dimensions.

Distinct descriptor-response rows could therefore be treated as

duplicates.

The scientific outputs were not rewritten.

The comparator was corrected through a named successor version.

The corrected comparator passed 50 tests and was applied read-only to

the immutable fresh and replay bundles.

This episode matters for two reasons.

First,

the initial comparison result could not be trusted until the identity

key was repaired.

Second,

the defect did not justify rerunning or silently replacing the

scientific execution.

The comparison instrument was corrected,

then applied again to the preserved evidence.

Governing rule

A defect in the adjudication layer must not be repaired by rewriting
the scientific evidence it was meant to inspect.
Correct the comparator.
Preserve the run.
Re-adjudicate read-only.

4.11 Replay result

The final comparison included:

- 192 corresponding files;
- 28,248 numerical cells.

The result was:

Table 4.5: JARVIS fresh-versus-replay adjudication.

Contract	**Result**	**Meaning**
Ordered plan identity	Pass	The same 48-unit scientific plan was completed.
Semantic identity	Pass	Corresponding values referred to the same scientific objects.
Numerical equivalence	Pass	All 28,248 compared numeric cells remained within 10^{-10}.
Maximum absolute difference	1.07×10^{-14}	Maximum observed numerical divergence.
Raw-byte identity	Fail for 60 of 192 files	Scientific values reproduced, but file bytes were not universally identical.

The correct terminal description is:

> Scientific replay equivalent,
>
> byte divergent.

The result does not establish cross-platform reproducibility.

It establishes local locked replay under the frozen environment and

tolerance.

4.12 Runtime and resource accessibility

The JARVIS fresh execution completed in approximately:

$$9.93 \text{ hours.} \tag{4.11}$$

It used:

- eight bounded workers;
- a consumer workstation;
- 32 GB of installed RAM;
- no GPU dependency for the tabular execution.

The underlying JARVIS DFT calculations were not performed locally.

The runtime refers to the Geatomica modelling,

perturbation,

null,

removal

and evidence-generation workload over published records.

The machine remained available for ordinary use during much of the

run.

This is relevant to practical accessibility,

but it is not a formal benchmark.

The defensible conclusion is:

> A bounded full-scale materials assurance workload completed locally on ordinary workstation hardware without requiring dedicated accelerator or cloud infrastructure.

4.13 What the JARVIS stage established

The JARVIS stage supports the following bounded claims:

1. All four targets contained meaningful non-null predictive signal under the declared tabular routes.
2. Target-specific admissible information improved performance.
3. Larger populations improved prediction across all target/lane aggregates.
4. Model family materially changed the apparent explanatory structure.
5. Population splitting changed performance,
   and no universal hardest split was observed.
6. Numerical noise and descriptor dropout exposed different forms of sensitivity.
7. The descriptor account was target-dependent.
8. The complete scientific programme was reproduced by a predecessor-linked replay within the frozen numerical tolerance.
9. Numerical reproducibility did not imply raw-byte identity.
10. An interrupted replay could resume from verified scientific-unit boundaries.

4.14 What the JARVIS stage did not establish

The stage did not establish:

- physical causality;
- a universal descriptor hierarchy;

- material-by-material transfer;
- cross-source invariance;
- experimental validity;
- physical material-selection authority;
- universal zone thresholds;
- a completed reduced-complexity descriptor spine.

JARVIS claim ceiling

The JARVIS programme established a coherent,
non-null,
target-dependent and numerically replayable descriptor-response
surface within one governed calculated source environment.
It did not establish that the same response surface would survive a
second source.
That question required the Materials Project experiment.

4.15 Chapter conclusion

JARVIS demonstrated that Geatomica could operate at meaningful source

scale and return more than one predictive score.

The experiment separated:

- target-dependent carrier signal;
- scale behaviour;
- population-transfer limits;
- model-conditioned explanation;
- noise and missingness sensitivity;
- scientific numerical replay;
- raw-byte divergence.

The result was promising,

but it remained source-local.

The next programme stage therefore changed the source environment.

That stage did not produce a simple clean validation.

It exposed the more difficult problem Geatomica had been built to

find.

Chapter 5

Materials Project: When Predictive Signal Meets Unresolved Evidence

Question. Can a route beat a paired broken-target comparator while still failing a minimal absolute predictive baseline?
Comparison. For the same 288 eligible elastic cells, paired-null separation was compared with held-out $R^2 > 0$.
Result. 211/288 (73.3%) beat the paired null, but only 13/288 (4.51%) achieved positive R^2 and passed both; median route R^2 was -2.248.
Decision. Require non-null signal and absolute viability separately; hold unresolved elastic source semantics.
Boundary. The $R^2 > 0$ boundary is the minimal demonstrated reference for this audit, not a universal engineering threshold.

5.1 Why move to a second source?

The JARVIS programme established that Geatomica could execute a

complete four-target audit,

retain target-dependent behaviour

and reproduce the scientific response surface under a locked replay.

That result was necessary.

It was not sufficient.

A descriptor account that survives only inside one calculated database

may reflect:

- the material population selected by that source;
- the source's calculation methods;
- its convergence and admission rules;
- its reporting conventions;
- its available descriptor fields;
- its historical database construction.

If descriptor authority was to mean more than source-local stability,

the audit had to enter a second computational environment.

The Materials Project therefore served two roles.

First,

it tested whether energetic and electronic descriptor behaviour would

recur across a separate source population.

Second,

it tested whether the assurance logic could survive a less cooperative

result.

The first role produced bounded support.

The second changed the programme.

Governing rule

A second source is not valuable only when it confirms the first. It is equally valuable when it reveals that the original evidence contract was incomplete.

5.2 The Materials Project programme

The Materials Project stage retained the four-property structure:

- formation energy;
- band gap;
- bulk modulus;
- shear modulus.

The programme expanded to:

$$64 \tag{5.1}$$

fresh scientific units and:

$$64 \tag{5.2}$$

replay units.

The source universe contained approximately:

$$154{,}377 \tag{5.3}$$

unique material records,

with target-specific cohorts constructed according to field
availability and the frozen modelling contract.
Together,
the JARVIS and Materials Project programmes touched approximately:

$$230{,}370 \tag{5.4}$$

calculated source records while retaining the two databases as
separate evidence environments.
The programme did not merge JARVIS and Materials Project into one
homogeneous table.
That separation matters.
A combined table would increase sample size while obscuring which
behaviour belonged to which source environment.

Table 5.1: Role and outcome of the Materials Project programme.

Object	**Observed value**	**Interpretation**
Approximate source universe	154,377 unique records	Materials Project source population used by the governed build.
Fresh scientific units	64/64	Complete execution.
Replay scientific units	64/64	Complete replay under a separate execution identity.
Formation-energy route	Bounded support	Predictive and descriptor behaviour remained coherent under the declared source and split conditions.
Band-gap route	Bounded support	Cross-source energetic/electronic recurrence was observed at aggregate level.
Raw elastic source	Source hold	Extreme and unresolved values prevented physical interpretation.
Elastic numerical replay	Replay hold	Completed, but exceeded the frozen absolute tolerance in 916 numeric cells.

5.3 Energetic and electronic recurrence

Formation energy and band gap produced coherent bounded results.
At full scale,

median R^2 increased from:

$$0.675 \rightarrow 0.747 \tag{5.5}$$

for formation energy when target-specific admissible descriptors were

added.

For band gap,

the corresponding increase was:

$$0.559 \rightarrow 0.719. \tag{5.6}$$

Signal persisted under composition and chemical-system holdouts,

although performance attenuated.

This broadly recurred with the JARVIS energetic and electronic pattern:

- target-specific information increased recoverable signal;
- harder population splits reduced performance;
- the descriptor account remained conditional on source,

  model

  and population geometry;
- energetic and electronic targets remained more coherent than the raw

  elastic routes.

The result supports cross-source recurrence at the population level.

It does not establish matched-material transfer.

Only nine exact structure fingerprints overlapped under the strict

matching route.

The two sources therefore act as provenance-separated witnesses,

not as two measurements of the same large set of materials.

Cross-source recurrence boundary

The Materials Project and JARVIS results support a bounded statement that similar energetic and electronic modelling behaviour recurred across two independently governed calculated source environments. They do not establish that an individual material carries the same descriptor response across both sources.

5.4 Availability is not admissibility

The elastic route exposed a more fundamental problem.

The source contained populated elastic fields.

That did not establish that every value was physically or numerically

fit for interpretation.

Before a model can answer what an elastic population means,

the source population must first establish:

- material identity;
- property definition;
- units;
- calculation method;
- tensor or scalar status;
- sign meaning;
- Voigt,

  Reuss

  and Hill consistency;
- physical bounds;
- whether extreme values represent unstable structures,

  failed calculations,

  parsing defects

  or sentinel values.

The original modelling route had treated field availability as

sufficient for intake.

The experiment showed that this assumption was too weak.

The source itself became the object of study.

Engineering consequence

An engineer cannot interpret a model's elastic ranking before deciding
whether the target population is physically and numerically admissible.
The model cannot repair the meaning of the source after fitting.

5.5 The elastic-source anomalies and semantic uncertainty

The raw elastic routes exposed values at extreme scales.

Reported bulk-modulus values extended to approximately:

$$-4.70 \times 10^{11} \text{ GPa}, \tag{5.7}$$

while reported shear values extended from approximately:

$$-1.43 \times 10^{13} \text{ GPa} \tag{5.8}$$

to:

$$5.65 \times 10^{13} \text{ GPa}. \tag{5.9}$$

Negative elastic quantities are not automatically database errors.

A mechanically unstable theoretical structure can produce negative elastic behaviour.

A simple sign filter would therefore destroy potentially meaningful physics.

The issue was the combination of:

- extreme magnitude;
- uncertain source semantics;
- mixed sign behaviour;
- bound-order inconsistencies;
- large influence on prediction and replay.

These values dominated:

- loss functions;
- tree partitions;
- model comparisons;
- feature rankings;
- numerical replay sensitivity.

A model trained on this population could still learn structure.

That did not mean the resulting structure was physically interpretable.

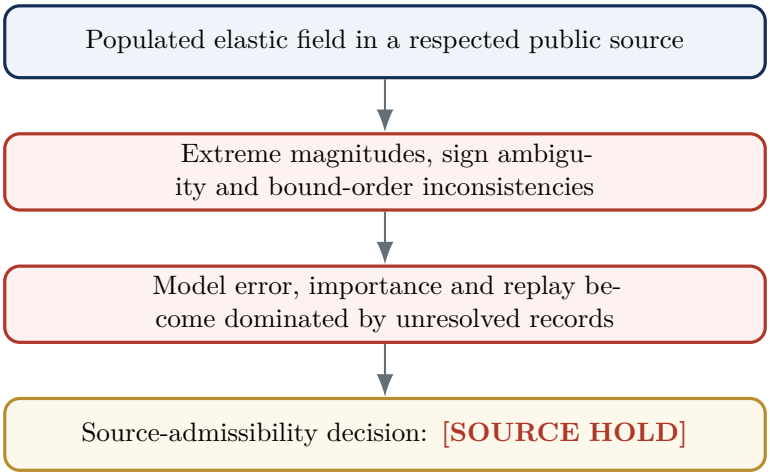


Figure 5.1: The elastic failure occurred before interpretation. A populated source field entered the modelling route, but unresolved physical and numerical semantics prevented the resulting model evidence from acquiring authority.

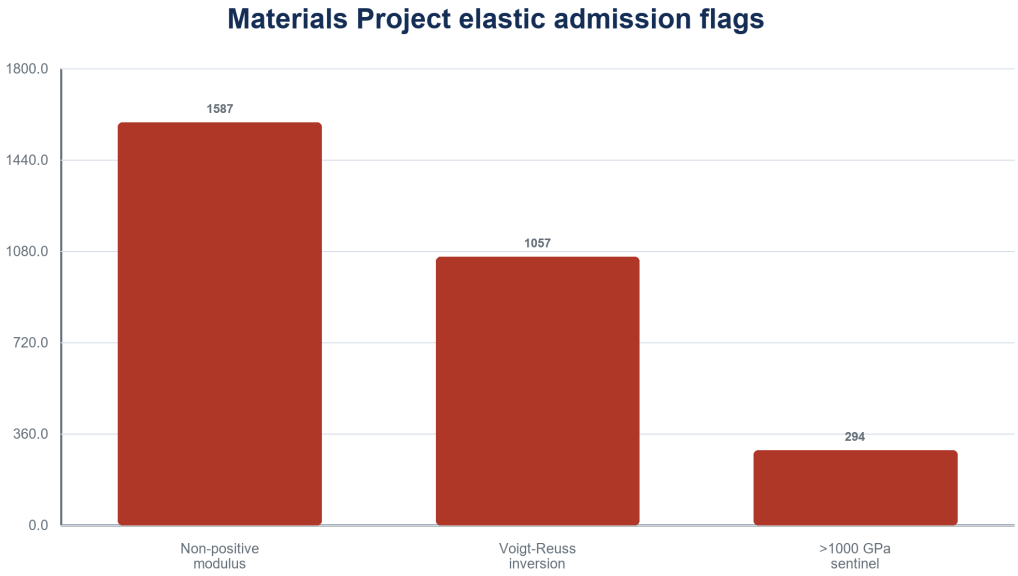


Figure 5.2: **HISTORICAL REPORT BOUND; SOURCE HOLD.** Elastic admission flags reported for the Materials Project route. Classes can overlap and begin adjudication; they do not prove that every flagged record is corrupt.

5.6 The prospective no-fit admission audit

A successor no-fit admission classifier evaluated:

$$26{,}056 \tag{5.10}$$

elastic record-target pairs.

The audit classified:

$$2{,}165 \tag{5.11}$$

pairs as anomalous and unresolved under the declared rules.

Observed anomaly classes included:

- 1,587 non-positive scalar cases;
- 1,057 Voigt/Reuss order reversals;
- 294 values above a

  $1000\,\mathrm{GPa}$

  review sentinel.

These classes overlap.

Their counts should not be added as mutually exclusive categories.

The audit also identified an example associated with

$\mathrm{TiCo_2}$

at an extreme reported magnitude of approximately:

$$-9.40 \times 10^{11}\ \mathrm{GPa}. \tag{5.12}$$

The no-fit classifier did not claim that every flagged record was scientifically invalid.

It separated records that could not proceed directly into ordinary interpretation without source resolution.

Table 5.2: Prospective admission classification for the Materials Project elastic record-target pairs.

Admission class	Count	Meaning
ADMISSIBLE_CONDITIONAL	23,891	Scalar property present, but complete tensor and record-level method context were not sufficient for unconditional physical interpretation.
ANOMALOUS_UNRESOLVED	2,165	At least one declared sign, bound-order or extreme-value rule required resolution before modelling authority.
ADMISSIBLE_STABLE	0	No record met the stronger stable-admission condition under the available source evidence.

The most important change was procedural.

The audit moved admission before fitting.

A later model could no longer legitimise an earlier unresolved source

population merely by producing a useful score.

Governing rule

Availability answers whether a field exists.
Admission answers whether that field is fit for the scientific claim being requested.
These are separate gates.

5.7 The database was not the failure

The result should not be interpreted as a claim that the Materials

Project is a failed database.

Large public scientific databases necessarily contain:

- edge cases;
- unstable calculated structures;
- heterogeneous historical records;
- evolving calculation methods;
- partial fields;
- legacy conventions;
- values whose meaning depends on source-specific context.

These conditions are part of real scientific infrastructure.

The failure would be to treat a large public database as if every

populated field formed one immediately homogeneous and physically

interpretable population.

The source provided evidence.

The modelling route had to decide what kind of evidence it was.

5.8 The elasticity paradox

The elastic route exposed another failure that was independent of

source admission.

Across the raw untransformed elasticity programme:

$$88.0\% \tag{5.13}$$

of model cells achieved positive valid-minus-null separation:

This 88.0% figure describes the earlier historical raw-elasticity audit under its own eligibility boundary. It is not the same-denominator analysis in Paper 2, where 211 of 288 cells (73.3%) had positive paired-null separation and 13 of 288 (4.51%) had positive held-out R^2.

$$S > 0. \tag{5.14}$$

On that basis,

the original aggregation rule treated much of the route as supported.

However,

median baseline predictive performance remained negative:

$$R^2 < 0. \tag{5.15}$$

The valid model had learned more than the paired target-permutation

null.

It had not learned enough to outperform prediction of the dataset

mean.

The same model result therefore supported two statements:

1. non-random signal exists under the declared comparison;
2. the resulting model is still absolutely unusable.

The first statement did not cancel the second.

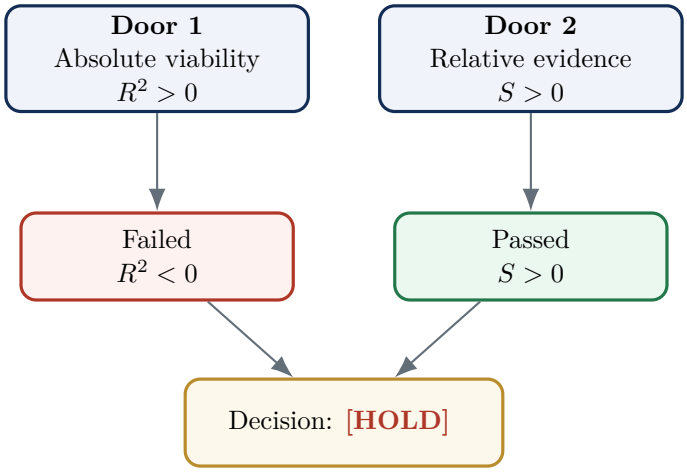


Figure 5.3: The two-door model-viability gate. Positive null separation cannot repair failed absolute prediction. Both doors must pass before bounded interpretation is nominated.

5.9 Why the paradox matters physically

A relative-only pass can still change an engineering decision if it is

reported carelessly.

Imagine a model used to prioritise candidate materials for:

- additional DFT calculation;
- synthesis;
- mechanical testing;
- high-temperature exposure;
- microscopy;
- neutron or synchrotron characterisation;
- prototype fabrication.

If the model beats a deliberately weakened null,

the result may be described as evidence that the screening route works.

If the model remains worse than predicting the mean,

that description is not enough.

The engineer may still spend:

- compute time;
- laboratory time;
- material stock;
- machine access;

- staff effort;
- experimental budget;
- calendar time.

The two-door rule therefore changes what is allowed to happen next.

It is not merely a change in terminology.

Engineering consequence

A model that passes only the relative-evidence door may justify further diagnosis.
It does not justify engineering use.

5.10 The successor viability rule

The corrected rule requires:

$$\text{viable} \iff \Big(R^2 > 0\Big) \wedge (S > 0)\,. \tag{5.16}$$

The original aggregation was preserved as historical evidence.

The corrected rule was introduced under a successor protocol.

The result was not rewritten to imply that the gate had always existed.

This matters because the programme itself is part of the evidence.

A scientific method that changes after observing a failure must retain

the sequence:

1. original rule;
2. observed defect;
3. preserved historical output;
4. successor rule;
5. new claim boundary.

5.11 Fresh execution and replay

The Materials Project fresh execution completed:

$$64/64 \tag{5.17}$$

scientific units.

The replay also completed:

$$64/64 \tag{5.18}$$

units.

The numerical comparison covered:

$$37{,}924 \tag{5.19}$$

cells.

Of these:

$$916 \tag{5.20}$$

exceeded the frozen absolute tolerance of:

$$10^{-10}. \tag{5.21}$$

Every exceedance occurred in bulk- or shear-modulus routes.

Formation energy and band gap remained within the tolerance.

Table 5.3: Materials Project replay outcome by evidence route.

Route	**Replay state**	**Interpretation**
Formation energy	Numerical pass	Fresh and replay values remained within the frozen tolerance.
Band gap	Numerical pass	Fresh and replay values remained within the frozen tolerance.
Bulk modulus	Replay hold	At least part of the compared response exceeded the frozen absolute tolerance.
Shear modulus	Replay hold	At least part of the compared response exceeded the frozen absolute tolerance.

The elastic replay still preserved:

- model-family ordering;
- leading descriptor ordering;
- null-separation signs;
- baseline-sign conclusions.

The scientific pattern recurred.

The strict numerical contract failed.

Both results were retained.

Governing rule

A stable scientific ordering does not repair a failed numerical replay contract.
A failed numerical replay contract does not erase every repeated scientific pattern.
Mixed evidence must remain mixed.

5.12 Why the tolerance was not widened

After observing the replay divergence,

it would have been possible to argue that an absolute tolerance of:

$$10^{-10} \tag{5.22}$$

was unnecessarily strict for targets spanning extreme magnitudes.

That may be true.

It does not justify changing the predeclared contract after seeing

the result.

The correct procedure is:

1. retain the original failed adjudication;
2. diagnose whether absolute,

   relative

   or scale-normalised tolerance is more appropriate;
3. define a successor replay contract;
4. execute the successor under a new identity;
5. compare the conclusions without rewriting the original result.

The Materials Project elastic replay therefore remains held under the

original contract.

5.13 What changed during the Materials Project stage

The Materials Project experiment changed the programme in five ways.

5.13.1 Admission moved before fitting

A populated source field could no longer enter ordinary interpretation

without a declared admissibility decision.

5.13.2 Relative evidence was separated from absolute viability

Positive null separation could no longer authorise an unusable model.

5.13.3 Mixed states replaced one terminal success label

Formation energy,

band gap,

bulk modulus

and shear modulus could return different evidence states inside the

same completed programme.

5.13.4 Reproducibility became multi-contract

Semantic recurrence,

numerical equivalence

and byte identity were treated separately.

5.13.5 A negative result became an engineering output

The hold was not a failed attempt to find a positive story.

It was the correct terminal result for the available evidence.

Materials Project terminal evidence state

- Formation energy:
  bounded cross-source support and numerical replay pass.
- Band gap:
  bounded cross-source support and numerical replay pass.
- Raw elasticity:
  source-admissibility hold.
- Elastic strict replay:
  hold under the frozen absolute tolerance.
- Historical outputs:
  preserved.
- Successor protocol:
  required for any revised elastic admission or replay rule.

5.14 What the Materials Project stage established

The Materials Project stage supports the following bounded claims:

1. Energetic and electronic modelling behaviour recurred at aggregate

   level across JARVIS and Materials Project.

2. Cross-source recurrence did not eliminate split,

   model

   or representation dependence.

3. A populated elastic field was not automatically admissible for physical interpretation.
4. The prospective admission layer identified a substantial unresolved record class.
5. Positive separation from a paired null was insufficient for model viability.
6. The two-door rule was required to separate non-random signal from usable prediction.
7. Fresh and replay executions can preserve scientific ordering while failing a stricter numerical contract.
8. A completed scientific programme can return a valid hold.

5.15 What the Materials Project stage did not establish

The stage did not establish:

- that every flagged elastic record is erroneous;
- that negative modulus values are intrinsically invalid;
- that the Materials Project is an unreliable scientific source;
- that elasticity is inherently unpredictable;
- that cross-source recurrence implies matched-material agreement;
- that log transformation is universally correct;
- that the successor admission rules have been externally validated;
- that a held route can never be recovered through better source resolution.

Materials Project claim ceiling

The Materials Project stage reduced confidence in single-number evidence.
It did not reduce confidence in scientific modelling itself.
The result showed that predictive signal,
source meaning,
absolute viability
and replay equivalence must be adjudicated separately.

5.16 Chapter conclusion

The Materials Project stage did not provide a simpler repetition of

JARVIS.

It exposed the boundaries that the first complete audit could not reveal by itself.

The strongest result was not a new regressor.

It was a change in what counted as evidence.

A source field had to be admitted.

A model had to be absolutely useful.

A null had to fail in the expected direction.

A replay had to satisfy its own declared contract.

A mixed scientific programme had to be allowed to return a smaller honest answer.

The next stage entered a deliberately curated benchmark environment.

The question was no longer whether raw scientific evidence could fail.

It was whether strong benchmark performance would make the assurance problem disappear.

Chapter 6

Matbench: Sealed Fresh Assurance and Exact Repeat

Question. Does fitted permutation importance agree with complete retrained-removal consequence?
Comparison. Across four official Matbench tasks, 142 Magpie descriptors and five folds, fitted importance was compared with refitted individual and group removals under one frozen CPU-MLP protocol.
Result. Mean Spearman agreement was 0.039404 and mean top-10 overlap was 0.100; all 3,440 fresh and 3,440 repeat units matched exactly across 65,740 numeric cells.
Decision. Treat fitted use and retrained dependence as different measurements.
Boundary. The result is same-environment repeatability under one representation/model protocol, not causal necessity or external replication.

6.1 Evidence transition

The previous edition described Matbench V0.3 only as an active operational patch. The fresh execution has now reached **MEASURED LOCAL** terminal state: all 3,440 predeclared units completed and the final manifest sealed. The separately identified repeat completed all 3,440 units under a distinct execution identity. All 65,740 compared numeric cells were exactly equal and semantic identity passed for every unit. This chapter therefore treats same-environment repeatability as measured while withholding external replication.

The execution used four official Matbench regression tasks, five official folds, 142 composition-derived Magpie features, and a fixed scikit-learn `MLPRegressor(128,64)` route. It contained 20 baseline/perturbation units, 2,840 retrained individual-feature removals and 580 retrained property-group removals. accepted evidence memory was disabled and recorded zero promotions.

6.2 Predictive carrier

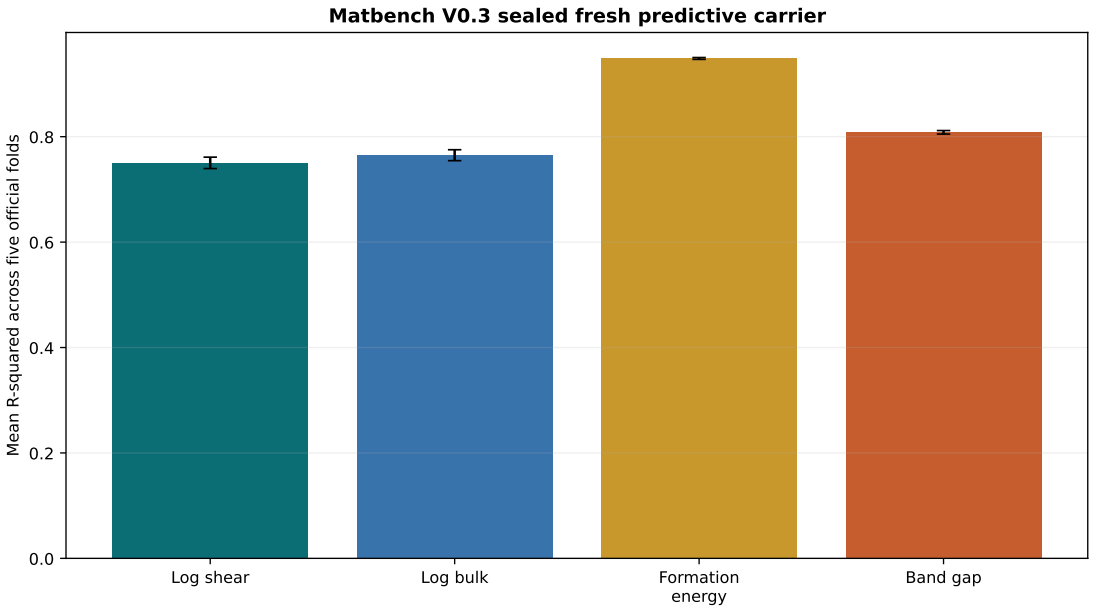


Figure 6.1: **MEASURED LOCAL FRESH RUN.** Mean baseline predictive carrier across five official folds. Error bars show fold standard deviation. Replay is completed and established exact same-environment repeatability.

Table 6.1: Matbench V0.3 sealed fresh baseline results. MAE remains task-native and must not be compared as one physical scale across targets.

Task	Mean R^2	SD R^2	Mean MAE
Formation energy	0.9484	0.0017	0.1447
Band gap	0.8083	0.0033	0.4245
Log bulk modulus	0.7649	0.0104	0.1202
Log shear modulus	0.7504	0.0108	0.1334

The fixed neural carrier remained strongest for formation energy, followed by band gap, logarithmic bulk modulus and logarithmic shear modulus. Two-percent test-feature noise caused comparatively small mean MAE changes, while ten-percent cell dropout caused much larger degradation in every task. The carrier therefore survives small continuous disturbance better than broad missingness under this declared route.

6.3 Importance is not retrained-removal consequence

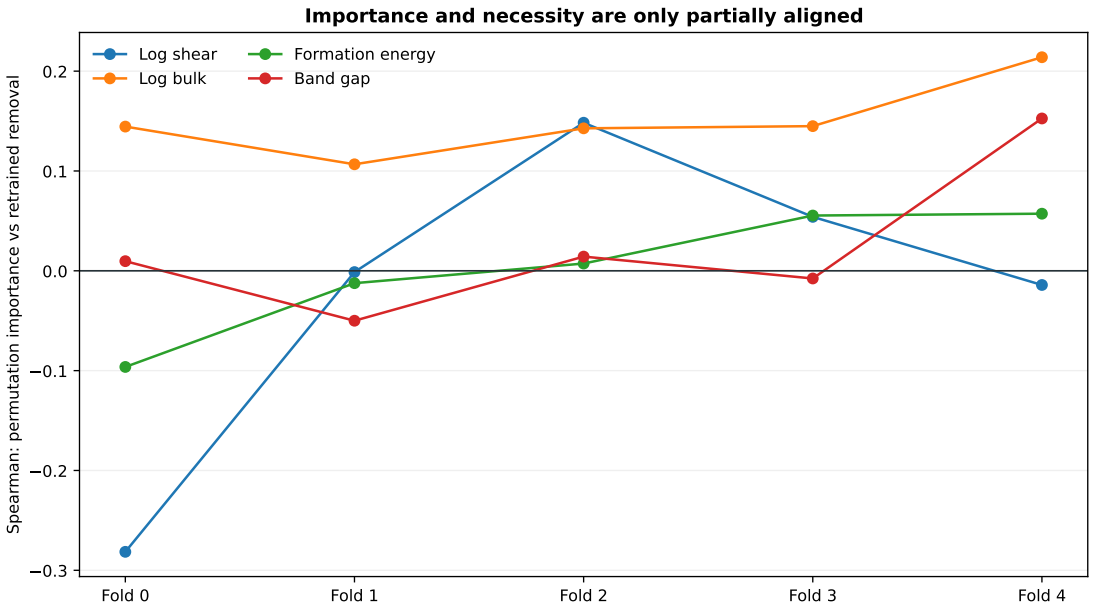


Figure 6.2: **MEASURED LOCAL FRESH RUN.** Fold-level Spearman association between fitted permutation importance and independently retrained single-feature removal consequence. Near-zero values indicate ranking disagreement, not causal irrelevance.

Across the 20 task-fold views, the mean Spearman association between fitted permutation MAE increase and retrained-removal MAE increase was 0.039, ranging from -0.282 to 0.214. The mean top-10 overlap was 10.0%

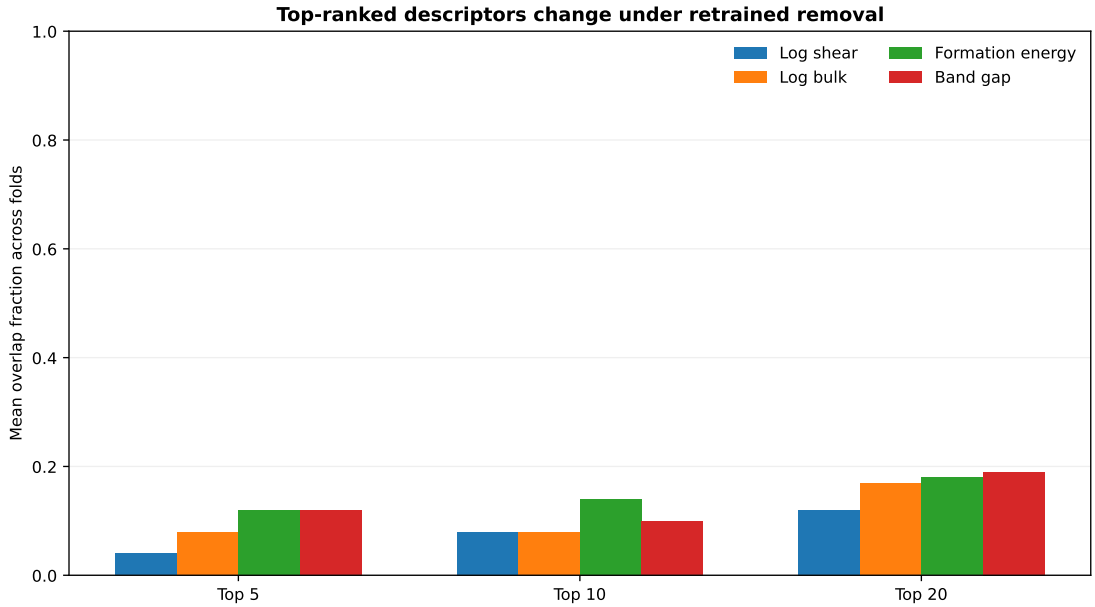


Figure 6.3: **MEASURED LOCAL FRESH RUN.** Mean overlap between descriptors ranked by fitted permutation response and descriptors ranked by retrained-removal consequence at $k = 5, 10, 20$.

The disagreement is not a subtle edge case. Of 2,840 individual removals, 1,781 improved MAE relative to the matched baseline and 1,059 worsened it. Of 580 group removals, 369 improved MAE and 211 worsened it. A feature can therefore be influential when disturbed in a fitted network without being necessary after the model is retrained and allowed to redistribute signal.

6.4 The necessary limitation

The removal route changes the input dimensionality and refits the MLP. Although the seed, fold and model contract are frozen, the current protocol contains no sham full-feature refit ensemble that directly measures optimisation variance without removing information. The observed disagreement can contain at least three components: descriptor substitutability, changed optimisation trajectory, and finite-sample variation. It is therefore evidence against treating fitted importance as necessity, but it is not a clean quantitative estimate of how much signal each descriptor uniquely contributes.

The next neural protocol should add repeated sham refits and matched random-column controls before promoting any descriptor-level authority classification.

6.5 Runtime and replay state

The fresh execution ran for approximately 19.16 hours on the local CPU workstation with two workers. Summed unit wall time was 37.94 hours, consistent with approximately two-way bounded concurrency. The repeat required 20.60 hours, used a new execution identity and did not reuse fresh outputs. The frozen comparator passed, so the scientific state is:

[PASS] `MEASURED LOCAL; REPLAYED SAME-ENVIRONMENT`

6.6 Claim boundary

Supported: a sealed 3,440-unit fresh execution showing strong predictive carrier signal and substantial disagreement between fitted perturbation rankings and retrained removal consequences under the declared Matbench tasks.

Not supported: external replication, unique descriptor dependence, causal feature claims, engineering qualification, external validation, or accepted evidence-memory promotion.

Chapter 7

Spatial GIL: Representation Persistence at 256 Materials

Question. Did an initially promising angle-aware representation persist when the cohort increased to 256 materials?
Comparison. Simple and repaired angle-aware structural representations were compared across deterministic folds and bulk/shear targets.
Result. The earlier angle advantage did not persist reliably at 256 materials.
Decision. Reject automatic scale-up or GNN escalation; diagnose representation blocks first.
Boundary. This is a bounded local persistence test, not proof that angular structure is universally unimportant.

7.1 The next representation question

The tabular programme asks whether an explanation survives disturbance in descriptor space. Spatial GIL moves the intervention toward crystal structure. The bounded V0.3 predecessor used 64 real JARVIS structures and found an apparent advantage for a repaired complete-shell angle-aware representation over a simple crystal-graph summary. The 256-material successor asked one narrower question:

> Does that angle-aware advantage persist across four times the sample size and four deterministic folds, and is any persistent gain larger for shear than bulk?

No GNN, new target, feature search or automatic 512-material escalation was authorised.

7.2 Protocol and execution

The result-blind cohort contains the first 256 ascending JARVIS identifiers with finite bulk and shear labels and a well-formed structure. Four contiguous 64-material test folds were frozen over that order. Each fold used 192 training records, Ridge with $alpha = 10^{-}6$, and two representations:

- a nine-coordinate simple crystal-graph summary;
- the same summary plus six density-normalised complete-shell angle bins.

Translation, rotation, atom permutation and supercell controls tested representation preservation; a small coordinate jitter remained a declared nonphysical positional-sensitivity stress. All 16 fresh units and all 16 replay units sealed. Replay passed raw-byte, semantic-metadata and frozen-numerical identity. The maximum representation-preserving prediction delta was 6.82×10^{-13}, below the 10^{-10} gate. accepted evidence-memory promotions remained zero.

7.3 The persistence result

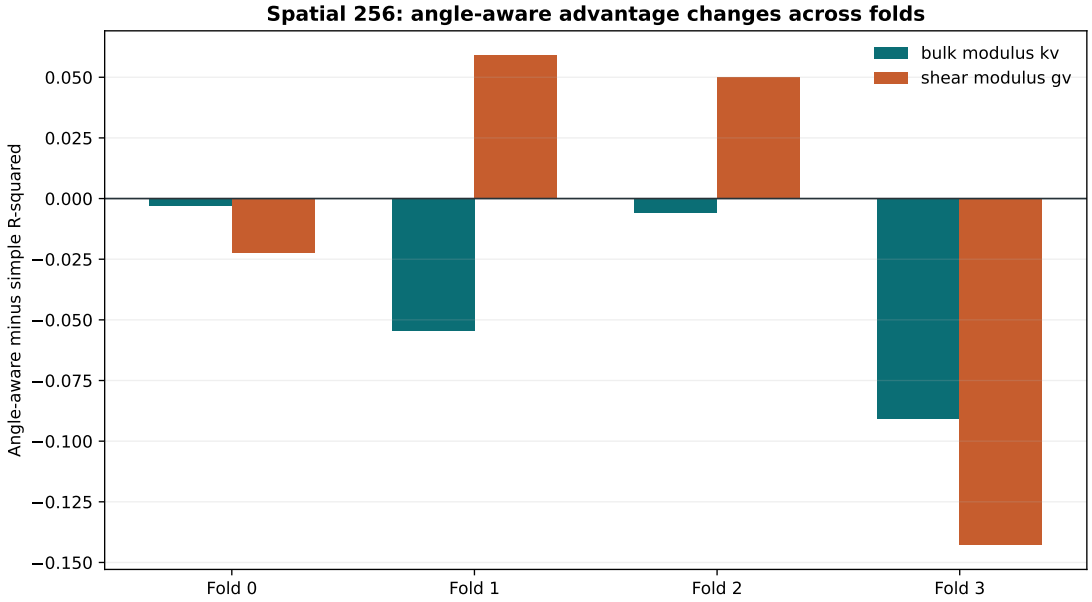


Figure 7.1: **SEALED AND REPLAYED LOCAL EVIDENCE.** Fold-level change in R^2 when the six complete-shell angle bins are added to the simple graph summary. Positive values favour the angle-aware representation.

The earlier 64-material direction did not persist. Bulk was worse with angle features in all four folds. Shear improved in two folds and worsened in two, including one large negative reversal.

Table 7.1: Pooled out-of-fold Spatial 256 performance.

Target	Simple R^2	Angle R^2	ΔR^2	MAE simple / angle
Bulk modulus	0.4447	0.4031	-0.0416	35.00 / 36.40
Shear modulus	0.3758	0.3513	-0.0244	20.69 / 21.26

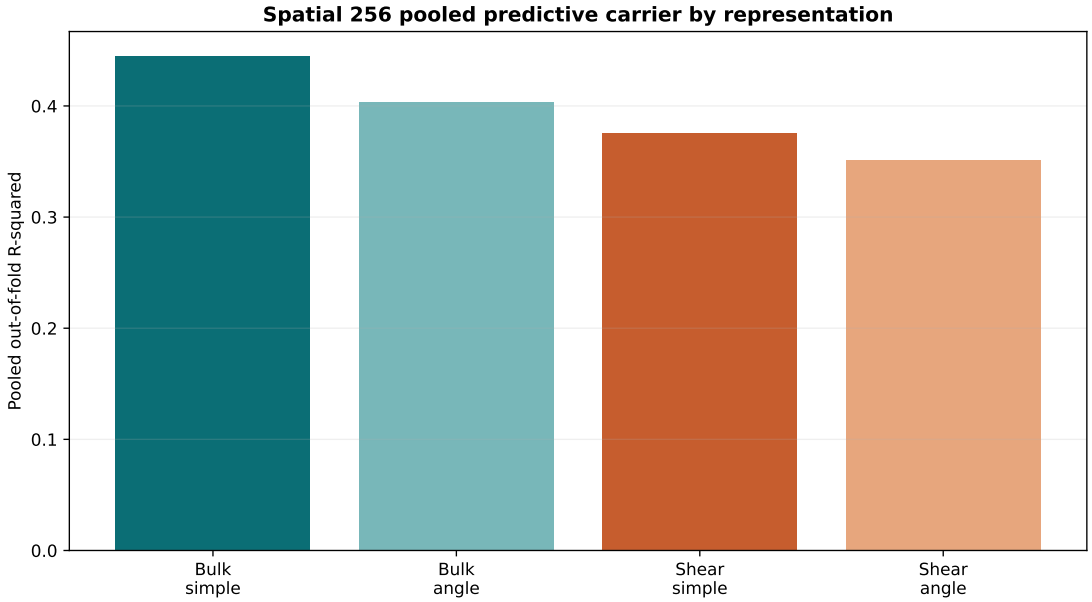


Figure 7.2: **SEALED AND REPLAYED LOCAL EVIDENCE.** Pooled out-of-fold carrier performance for the two representations and targets.

7.4 Why the angle summary failed to persist

The execution architecture passed; the hypothesis did not. The diagnostic identified three constraints.

1. The ascending-JID cohort and contiguous folds are deterministic and result-blind, but not chemistry-stratified. They test persistence across fixed blocks, not chemical transfer.
2. Six global histogram bins heavily compress local angular environments. Different structures can therefore map to similar angle summaries.
3. The density-normalised bins obey an effectively fixed-sum closure constraint. The simple standardised design condition number is 18.24; the angle matrix reaches approximately 3.75×10^{15} because one added direction is redundant. Post-hoc standardisation did not restore the angle advantage.

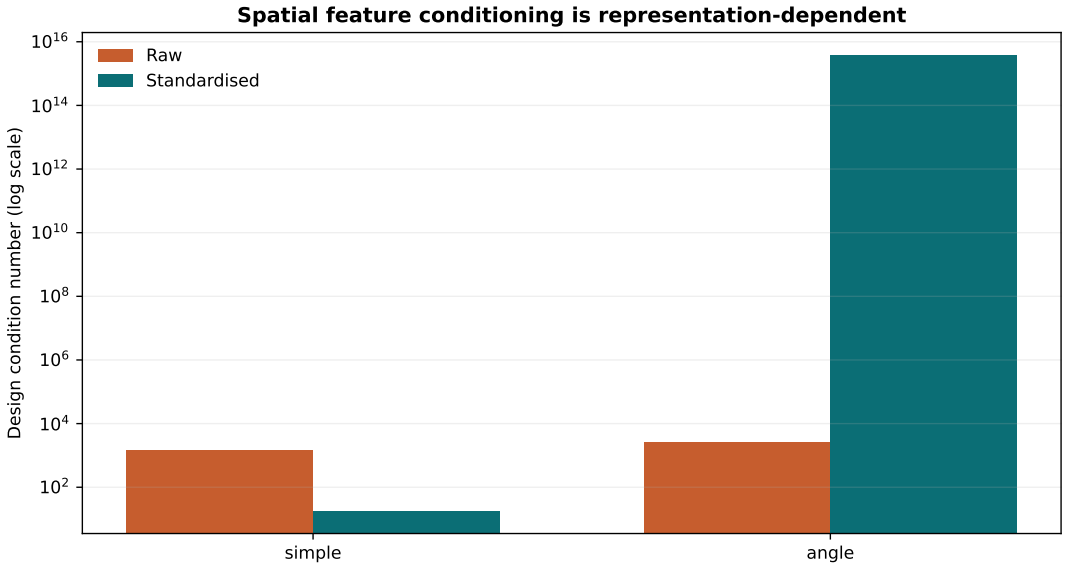


Figure 7.3: **POST-HOC DIAGNOSTIC.** Raw and standardised design condition numbers. This diagnostic explains a representation limitation but does not replace the sealed protocol result.

7.5 Decision

[HOLD] `512 NOT AUTOMATICALLY ADMITTED`

The next experiment should not merely increase the same sample rule. It should predeclare and freeze chemistry-aware memberships, remove the histogram closure redundancy, include a sham representation-expansion control, and test whether a less compressed angular representation earns its added cost. Only then should a 512-material stage be reconsidered.

7.6 Claim boundary

Supported: bounded local evidence that this repaired six-bin angle summary does not provide a persistent advantage over the simple graph summary for bulk or shear across the declared 256-material four-fold Ridge study.

Not supported: general inferiority of angular information, GNN conclusions, defect physics, materials recommendation, physical validation or external validation.

Chapter 8

Spatial GIL: Structural DOE, Confirmation and Exact Replay

Question. Which structural information persists after matched controls, population changes, and exact replay?
Comparison. A 32-configuration factorial study isolated representation blocks; V0.6.1 then tested bulk metric geometry and shear polarity across continuity and zero chemical-system-overlap views.
Result. Bulk metric geometry passed both views; shear polarity passed the chemical-system view but failed the frozen continuity threshold. All 927,361 replayed numeric cells matched exactly.
Decision. Confirm compact metric geometry for this bulk route; retain shear polarity as conditional.
Boundary. Frozen 256-material Ridge study only; no 512-material, GNN, causal, or physical-qualification authority.

8.1 Why decomposition followed persistence failure

The 64-material predecessor suggested an angle-aware advantage. At 256 materials the six-bin angular summary failed to retain a persistent benefit. The correct response was not immediate GNN escalation but a controlled 32-configuration decomposition of coordination, continuous metric geometry and polarity proxies.

8.2 V0.5 representation DOE

The factorial study identified continuous metric geometry as the strongest compact block: its mean R2 effect was +0.3503 for bulk and +0.2688 for shear. Coordination was harmful, generic angular geometry was harmful or unstable, polarity was conditional, and the full 43-feature representation underperformed selected smaller configurations. These were block-level results, not proof of individual-feature mechanism.

8.3 V0.6.1 confirmation contract

The successor froze 256 materials, Ridge, two targets, continuity and zero chemical-system-overlap views, four folds, nine configurations, matched shams and paired target-permutation nulls. It comprised 144 valid and 432 null units. Views could not repair each other by pooling.

8.4 Final measured and replayed result

Fresh and replay executions each sealed 576/576 units. All 1,728 corresponding output pairs were byte-identical; 927,361 numeric cells matched exactly with maximum difference zero; semantic

metadata and plan identity passed. Controls passed with maximum delta 2.331×10^{-12} against 10^{-8}. accepted evidence memory was disabled with zero promotions.

Bulk metric geometry exceeded BASE by +0.4855 R2 under zero chemical-system overlap and +0.4778 under continuity, exceeded matched shams by +0.5157 and +0.5145, and was positive in 4/4 folds in both views. It therefore confirmed. Shear polarity over metric geometry passed the grouped view (+0.0453; 4/4 folds) but failed continuity (+0.0123 against +0.02; 3/4 folds). It remains conditional.

Final spatial result

More representation is not automatically better representation. Metric geometry survived confirmation and exact replay; the stronger universal polarity claim did not.

8.5 Claim boundary

The result is measured and controller-replayed locally for the frozen cohort, model, folds, nulls and controls. It is not external validation, universal mechanism, 512- material authority, GNN superiority, physical qualification or deployment permission.

Chapter 9

Three-Paper Decomposition and Research Questions

The monograph is the synthesis object. It should not be submitted unchanged as three overlapping articles. The publication programme separates three claims that can be reviewed and falsified independently.

Article	Primary question	Decisive evidence	Excluded claim
Paper 1: Importance Is Not Necessity	Do fitted rankings agree with retrained removal consequences?	Matbench V0.3 sealed fresh, JARVIS bounded removal, and Spatial V0.5 block interventions	No external replication; no universal causal descriptor hierarchy
Paper 2: Better Than Null Is Not Yet Useful	Does a route satisfy both null separation and absolute/transfer viability?	JARVIS, held MP elasticity, Matbench task-native metrics, and Spatial split challenge	No invented null R^2; no external validation
Paper 3: Available Evidence Is Not Engineering Authority	Can evidence be admitted, replayed, and kept outside accepted memory when unsupported?	MP no-fit admissibility, recovery-controlled JARVIS and Spatial replay, zero-promotion evidence-memory controls	No database-wide corruption claim; no enterprise compliance claim

9.1 Open successors retained without result leakage

- **Matbench V0.3 repeat:** completed under a separate execution identity; 65,740 compared numeric cells were exactly equal.
- **Spatial GIL V0.6.1:** fresh and replay executions each sealed 576/576 units; exact replay retained bulk geometry PASS and shear polarity cross-view FAIL.
- **MP tensor/method admission:** held until tensor-bearing evidence, method receipts, domain review, and frozen cohort membership exist.
- **Public research compendium:** sanitised papers, evidence summaries, reference sources and contract boundaries prepared for archival deposit.

The queue is operational context, not a fourth evidence class. Every paper must be updated only after a queued item seals and its claims are adjudicated.

Chapter 10

What Changed During the Programme

10.1 Programme-level synthesis

Across the programme, the recurring failure mode was substitution between evidence objects: fitted use for retrained dependence, null separation for viability, availability for admission, replay for authority, or narrative memory for accepted scientific state. Geatomica therefore treats evidence as a stateful object whose authority increases only through explicit transitions.

Authority inheritance

A weaker state does not inherit the authority of a stronger one merely because both occur in the same workflow.

10.2 The framework was not the starting point

It is possible to read the completed architecture backwards and assume

that Geatomica began as a fully designed assurance framework.

It did not.

The programme began with a materials question:

> Can a smaller set of governing descriptors recover enough material
>
> behaviour to reduce modelling complexity without abandoning the
>
> underlying physics?

The later concepts of admission,

viability,

replay,

authority

and governance emerged because the original question repeatedly met

evidence that could not be interpreted safely through ordinary

single-score workflows.

The architecture is therefore best understood as the accumulated

response to experimental failure.

Each layer exists because an earlier layer proved insufficient.

10.3 First transition: from descriptor discovery to descriptor stability

The first ambition was descriptor discovery.

The programme sought compact relationships among:

- composition;
- electronic state;
- energetic organisation;
- structure;
- packing;
- mechanical response.

Initial models produced feature rankings.

Those rankings were promising,

but they were conditional.

Changing the model family,

population scale

or split geometry altered the apparent descriptor account.

The first transition was therefore:

$$\text{descriptor discovery} \rightarrow \text{descriptor stability}. \tag{10.1}$$

The scientific question changed from:

> Which descriptors rank highest?

to:

> Which descriptor relations remain difficult to remove when the
>
> modelling route changes?

10.4 Second transition: from stability to retrained dependence

Stability across seeds or perturbations did not establish that the

descriptor was load-bearing.

The bounded JARVIS Extra Trees removal lane showed why, while the sealed Matbench fresh and repeat executions expose a large, exactly repeated fitted-importance versus retrained-removal disagreement.

A fitted route can rank a descriptor highly while a retrained route reallocates signal through correlated alternatives. The Matbench result is measured and repeated within the frozen same-environment protocol; mechanism and external replication remain unresolved.

The second transition was:

$$\text{stability} \rightarrow \text{retrained-dependence testing.} \tag{10.2}$$

The question changed from:

> Does this descriptor keep appearing?

to:

> What happens when this descriptor is made unavailable and the model is allowed to reconstruct the problem?

10.5 Third transition: from signal to viability

The Materials Project elasticity route exposed a different weakness.

A model could beat its paired null and still remain worse than prediction of the mean.

The existence of non-random signal did not establish practical model usefulness.

The third transition was:

$$\text{signal detection} \rightarrow \text{model viability.} \tag{10.3}$$

The question changed from:

> Did the model learn more than nonsense?

to:

> Did the model learn enough to justify interpretation?

10.6 Fourth transition: from available data to admissible evidence

The raw Materials Project elastic population showed that a respected database can expose values that require context before modelling.

The fourth transition was:

$$\text{data availability} \rightarrow \text{scientific admission.} \tag{10.4}$$

The question changed from:

> Is the target field populated?

to:

Do the source identity,

units,

method,

sign

and physical bounds support the claim being requested?

10.7 Fifth transition: from repeatability to reproducibility contracts

The JARVIS replay reproduced all compared numerical cells within the

frozen tolerance while failing raw-byte identity.

The Materials Project replay preserved scientific ordering while

failing the strict elastic numerical contract.

The fifth transition was:

$$\text{repeatability} \rightarrow \text{multi-contract reproducibility.} \tag{10.5}$$

The question changed from:

Did the run happen again?

to:

Which parts of the run were reproduced:

plan,

semantics,

numbers,

ordering

or bytes?

10.8 Sixth transition: from successful execution to recoverable science

Long-running programmes were interrupted.

A normal modelling script would often restart from the beginning or

continue from an informal partial state.

The recovery controller instead used complete scientific units,

expected-output contracts

and verified checkpoints.

The sixth transition was:

$$\text{completed execution} \rightarrow \text{recoverable execution}. \tag{10.6}$$

The question changed from:

> Can this experiment finish?

to:

> Can this experiment survive interruption without changing what counts as accepted evidence?

10.9 Seventh transition: from evidence to authority

The programme eventually contained several valid but incompatible evidence states:

- strong prediction with replaceable descriptors;
- positive null separation with negative absolute performance;
- source availability without physical admissibility;
- scientific recurrence without strict numerical replay;
- numerical replay without byte identity;
- cross-source recurrence without matched-material transfer.

No single metric could represent these states honestly.

The seventh transition was:

$$\text{evidence} \rightarrow \text{bounded authority}. \tag{10.7}$$

The final question became:

> What decision,
>
> if any,
>
> does this exact combination of evidence permit?

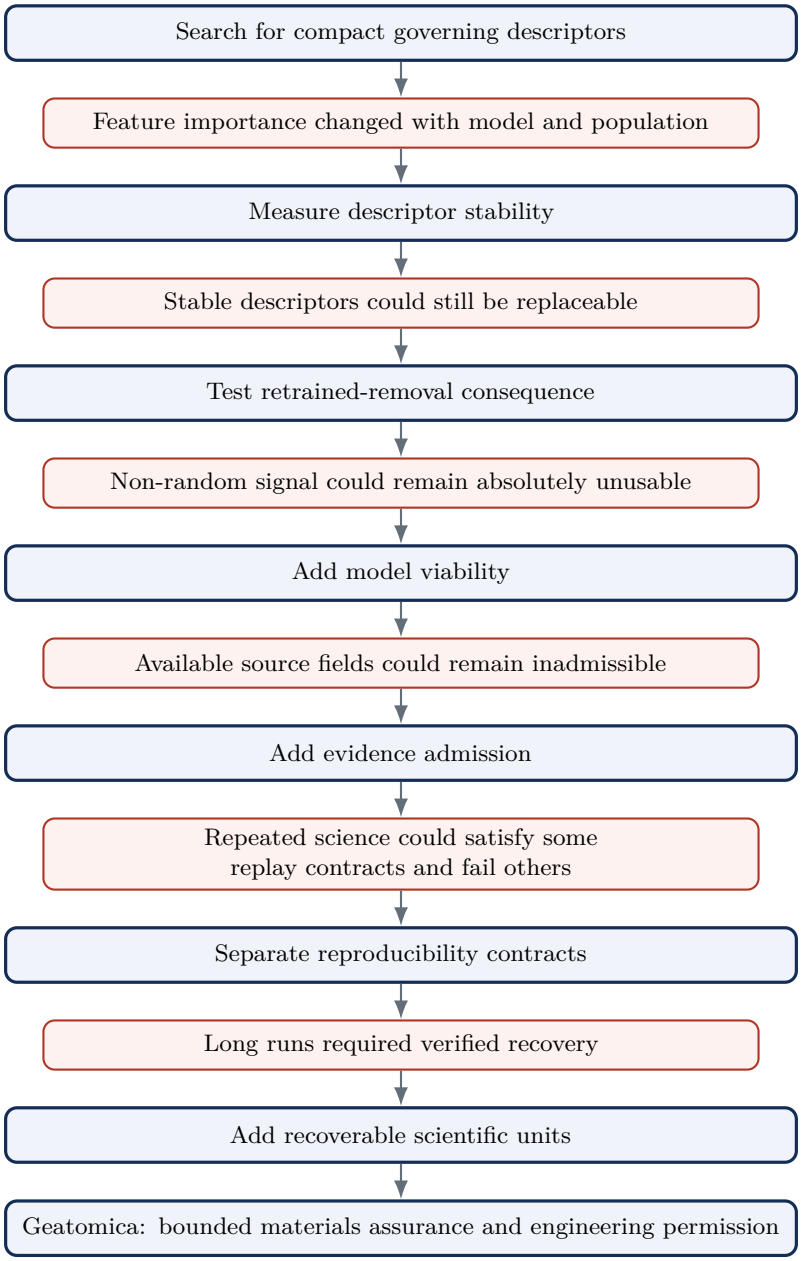


Figure 10.1: How the Geatomica programme changed. Each new layer was introduced after the preceding evidence object failed to support the decision being requested.

10.10 The unifying result: materials assurance

The completed programme is not best described as:

- a feature-selection method;
- a new regressor;
- a database-cleaning script;

- a reproducibility tool;
- a software-governance framework.

Each of those descriptions captures part of the work.

None captures the whole programme.

The unifying research object is

materials assurance.

Materials assurance asks:

> When does a result extracted from materials data deserve to influence an engineering decision?

That question sits between:

$$\text{materials} \rightarrow \text{evidence} \rightarrow \text{decision}. \tag{10.8}$$

The computational architecture is instrumentation around that chain.

10.11 Why this is not simply uncertainty quantification

Uncertainty quantification commonly estimates uncertainty in:

- predictions;
- parameters;
- data;
- model ensembles;
- distribution shift.

Geatomica overlaps with these concerns.

Its focus is different.

The programme asks whether different evidence objects agree on the permission being requested.

A model may show low predictive uncertainty while:

- using an inadmissible source population;
- depending on a replaceable descriptor;
- failing an applicability-domain transfer;
- failing strict replay;
- remaining absolutely worse than the mean.

Uncertainty estimates remain valuable.

They do not replace evidence adjudication.

10.12 Why this is not simply applicability domain

An applicability domain asks where a model's predictions are expected to remain reliable.

Descriptor authority asks an additional question:

> Does the explanation remain reliable enough to support the same engineering interpretation within that domain?

A model can predict well inside its applicability domain while using different correlated routes across learners.

Prediction domain and explanation domain are related.

They are not identical.

10.13 Why this is not simply feature stability

Feature stability asks whether a selected feature or ranking recurs across repeated fits or perturbed samples.

Descriptor authority also requires attention to:

- absolute model viability;
- negative controls;
- source admission;
- retrained removal;
- population transfer;
- replay;
- the action being requested.

A stable feature inside a failed carrier model does not gain engineering authority through repetition alone.

10.14 Why adverse results strengthened the programme

A conventional publication strategy might have removed or minimised the routes that did not produce clean positive results.

That would have removed the evidence that most clearly demonstrates the purpose of the programme.

The adverse results established that Geatomica could:

- detect a defect in its own aggregation logic;

- preserve the original output;
- introduce a successor rule without rewriting history;
- hold a respected source field before physical interpretation;
- complete a replay while refusing strict numerical equivalence;
- retain useful energetic and electronic evidence alongside held elasticity;
- return a smaller honest answer rather than one inflated headline.

These are not side effects.

They are the empirical basis of the assurance claim.

Governing rule

A scientific assurance system is not demonstrated only by the evidence it promotes.
It is demonstrated by the evidence it refuses to overstate.

10.15 The current research object

The final Geatomica research object can be stated as:

> A governed materials-informatics instrument that measures how predictive relationships, descriptor explanations and source evidence respond to declared changes, then issues only the bounded engineering permission supported by the surviving evidence.

Its principal outputs are:

- predictive carrier-signal evidence;
- descriptor-response barcodes;
- retrained-removal consequence results;
- source-admission states;
- model-viability states;
- population-transfer boundaries;
- replay contracts;
- recovery receipts;
- bounded engineering permissions;

- explicit refusals.

10.16 What remains unresolved

The programme has not yet completed its original scientific ambition.

It has not produced a validated universal reduced descriptor spine.

It has instead constructed the assurance conditions under which such a spine could be tested honestly.

The remaining questions include:

- Do any descriptor combinations persist across additional independent sources?
- Can electronic descriptors such as band gap or density-of-states quantities remain informative after controlling for material class?
- Do spatial and graph representations preserve the same descriptor zones?
- Can reduced descriptor sets guide new DFT calculations more efficiently?
- Do the nominated relationships survive physical measurements?
- Can dynamic material transitions be modelled without importing static feature logic into time-dependent systems?
- How should authority thresholds change with engineering consequence?

The assurance framework is therefore not the end of Geatomica.

It is the experimental substrate required for the next materials questions.

10.17 Chapter conclusion

The programme began with a search for compact governing descriptors.

The experiments forced that question to widen.

Feature importance was not enough.

Stability was not enough.

Null separation was not enough.

Source availability was not enough.

One completed run was not enough.

Even successful replay was not one indivisible state.

The resulting framework did not emerge because more terminology was desired.

It emerged because each previous evidence object failed to support the engineering permission being asked of it.

Geatomica is therefore best understood as a materials-assurance programme.

Its purpose is not to make every model trustworthy.

Its purpose is to make the boundary of trust measurable.

Chapter 11

Beyond Materials ML: Engineering Transfer and Bounded Design Studies

Materials ML exposed the evidence problem in a controlled scientific setting. It showed that a model could appear important, non-null, reproducible, or complete without earning the stronger interpretation being asked of it. The engineering studies exposed the same control problem at the action boundary: geometry changes broke transfer of thermal evidence, invalid state transitions remained invalid even when numerically evaluable, and fault containment improved while thermal performance degraded.

MATERIALS ML
What does the model or evidence actually support?
↓
importance / viability / replay distinctions
↓
GENERAL COMPUTATIONAL ENGINEERING
What may the result legitimately change next?

The engineering cases show transferable control behaviour across three local computational engineering studies. They do not constitute retrospective validation of the complete modern Systemica architecture, physical validation, external validation, or industrial deployment.

11.1 Three local engineering witnesses

Case	Starting decision	Consequential correction or failure	Bounded outcome and lineage
Furnace viewport	Whether a thermal-interface concept was mature enough to fabricate or test.	Correcting 350–375 mm to 35–40 mm broke transfer of an inherited 59.2 °C result. Active cooling added a dependency and failure route.	Thermal screen plus uncertainty/fault scenarios led to a guarded opaque thermal-interface coupon. Fabrication and furnace operation were not authorised. Later services formalised case identity, predecessor transfer, failure retention, and bounded action.

Dual tank	Whether a reserve, feeder, recovery, and return topology could advance.	Fresh reserve, live feeder/recovery, and returned or consumed states had to remain distinct. Forbidden return transitions were rejected even when a numerical route could be evaluated.	A low-pressure non-potable tracer rig was the next test; deployment was not authorised. Later services formalised typed state, topology, transition validity, and provenance.
Heat shield	Whether segmentation improved a thermal-flow protection concept.	Segmentation improved fault containment while degrading thermal performance. Blocked-cell and contained-leak cases remained visible rather than being averaged away.	Matched cold-flow and heated coupons were specified; qualification and flight readiness were not authorised. Later services formalised competing-objective and failure-case retention.

Across the three studies, 13 candidate objects, five descendants, 116 scenarios, 9,150 parameter or uncertainty runs, and at least 9,380 model operations narrowed broad concepts to smaller falsifiable tests. The counts show bounded computational activity, not labour savings, physical qualification, or industrial validation.

11.2 What was actually engineered and modelled

The three engineering studies were bounded pre-experimental calculations, not architecture demonstrations wearing engineering labels. They began with physical design questions and ended with smaller physical tests.

Case	**Physical object**	**Native model**	**Principal failure discovered**	**Bounded next test**
Furnace	High-temperature viewport/thermal interface	Deterministic dominant-path thermal screening with uncertainty and fault sweeps	Corrected geometry invalidated inherited evidence; cooling created a dependency; edge/frame transfer remained unresolved	Guarded opaque thermal-interface coupon
Dual tank	Reserve/feeder/recovery topology	Lumped flow/state model with typed topology and transition validation	Recovered water had been collapsed into reserve; invalid return routes remained possible	Low-pressure non-potable tracer rig
Heat shield	Segmented transpiration/fault-containment coupon	Transient cell thermal network with Darcy-style flow and topology/fault checks	Maximum nominal cooling and fault localisation diverged	Matched cold-flow/heated blocked/leak coupons

These are bounded screening and pre-experimental engineering instruments, not physical validation of the proposed systems.

11.2.1 Furnace thermal enclosure and viewport interface

The engineered object was a brazing-furnace thermal enclosure and removable optical/thermal viewport interface intended to provide process visibility without allowing the optical path to dominate external heat transfer or compromise the surrounding thermal boundary. The corrected intended interface envelope was 35–40 mm, not the inherited 350–375 mm. The previously reported 59.2 °C cold-side result was retained historically but refused for current design authority because it belonged to the wrong geometry identity.

The executed bounded model used deterministic dominant-path thermal-resistance screening, candidate-specific heat paths, six model/fault scenarios, 9,000 deterministic uncertainty samples, and 9,120 model operations. Variables included emissivity, hot-side boundary, contact resistance, wall depth, seal/interface assumptions, aperture, cooling-loss behaviour, and unresolved edge/frame transfer. F-A was a minimum-complexity single-pane control; F-B separated functions with a passive double-pane/cavity; F-C tested active cooling; F-A1 and F-B1 were descendants created after weaknesses appeared.

The second pane/cavity improved the declared central cold-side path, and a smaller aperture improved the pressure screen. Active cooling improved nominal temperature while adding a failure dependency. Variation across 35–40 mm did not remove the dominant optical path in this lightweight model, and edge/frame transfer remained unresolved. The engineering value was the conversion of an incorrectly inherited geometry and broad idea into traceable candidate families, quantified uncertainty, explicit failure dependencies, and a guarded coupon targeting the unresolved paths. It was not a validated final viewport.

11.2.2 Dual-tank provenance and recovery topology

The physical object was a multi-reservoir water distribution and recovery topology separating clean make-up reserve, live feeder/recovery flow, unused return, consumed water, and contained/rejected states while buffering demand. The correction made R the isolated new-source reserve and F the live feeder/recovery tank. Only declared unused return could pass a quality gate into F; consumed water exited; excess remained contained. Source provenance, age/residence, return history, and gate history were kept distinct because equal level or pressure does not imply equal reuse history.

The executed model combined lumped flow/state simulation, typed topology, candidate-specific transition validation, 26 scenarios, and 70 parameter cases. Forbidden routes were rejected before execution. W-B2 scored 33/35 under the exploratory screen; W-B4 was generated after W-B2 failed the peak criterion and scored 34/35. Its worst service-margin failure improved from -0.266 to -0.090 under those assumptions. These are not final sizing values.

The engineering contribution was correction of the physical state topology. Recirculation could recondition water without making it new-source water; capacity and provenance integrity remained different objectives. Final size, hygiene, pressure losses, water quality, and component ratings remain unresolved. The next action was a transparent low-pressure non-potable tracer rig measuring provenance, residence, and fault containment.

11.2.3 Segmented transpiration heat-shield coupon

The engineered object was a segmented porous/transpiration thermal-management ground coupon in which coolant delivery, thermal response, and cell-level fault containment were competing objectives. The model coupled a two-node-per-cell transient thermal network, Darcy-style permeability/restriction flow, structural topology, and transition/fault-route validation. It executed 84 scenarios and 80 parameter cases across six candidate/control objects, generated two descendants, and rejected four unsupported route classes.

Thermal nodes approximated coupon heat storage and transfer; restricted flow represented the porous delivery path; segmentation tested whether blockage or leakage remained local. H-A1, the monolithic route, carried the strongest nominal thermal signal with a temperature-reduction range of 175.3–389.5 K. H-B1, the segmented descendant, traded thermal response for isolation and maintenance information; its range included −83.5 to 245.4 K. Segmentation was therefore not a free improvement.

The model did not resolve pore-scale convection, choking or compressibility, external film formation, spectral radiation, oxidation, erosion or ablation, detailed contacts, or flight aerothermodynamics. Its value was to produce a falsifiable matched-coupon sequence: no-flow control, simple thermal/permeability route, segmented fault-isolation route, blocked-cell case, and contained-leak case. No qualification or flight readiness followed.

The technical contribution here is the design synthesis and engineering reduction demonstrated within the programme, not a claim of literature-level novelty for every physical mechanism.

11.3 Architecture provenance

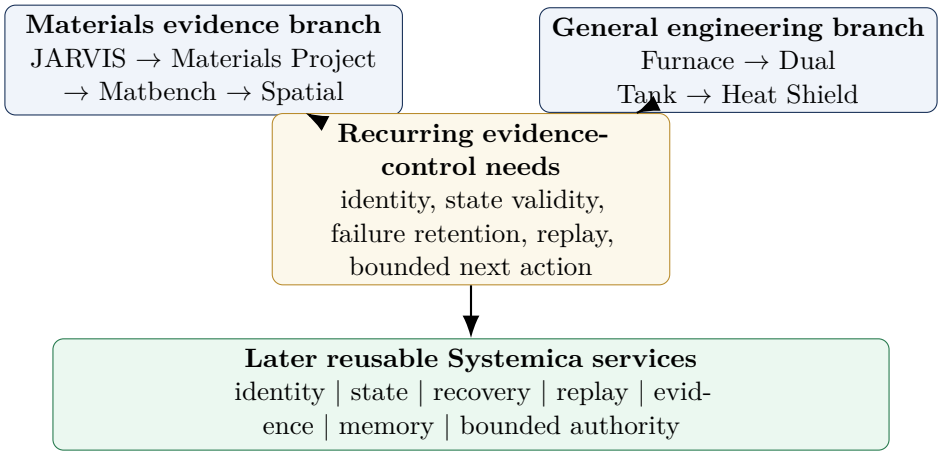


Figure 11.1: Two evidence branches converged on recurring control needs. The later architecture abstracts those behaviours; it was not retrospectively present in full in the historical cases.

Repeated behaviours observed across case-specific scientific and engineering workflows were progressively formalised into reusable execution, evidence, replay, memory, state, and bounded-authority services. Scientific-domain transfer remains unestablished beyond demonstrated materials semantics. Control-logic transfer has initial local evidence across materials ML/data analysis, thermal-interface design, fluid/state topology, and thermal-flow/fault-containment design.

11.4 Evidence becomes useful when it changes the next action

The preceding chapters established that materials-model evidence can separate into several distinct states.

A model can predict well while its explanation remains replaceable.

A relationship can recur across runs while failing under a harder population split.

A populated source field can remain unresolved for physical interpretation.

A model can beat a null while remaining worse than predicting the mean.

A scientific pattern can recur while a stricter numerical replay contract fails.

These distinctions become practically useful only when they alter what an engineer does next.

The purpose of Geatomica is therefore not to add another explanatory layer after prediction.

It is to place a decision boundary between a modelling result and the engineering action that may follow from it.

That boundary asks:

> What is the least expensive next action that remains scientifically justified by the surviving evidence?

The answer may be to continue using the current route.

It may be to escalate to a richer calculation or physical test.

It may be to request additional source information.

It may be to stop.

Governing rule

A model result should not acquire authority because it is available. It should acquire only the authority required for the next justified action.

11.5 The engineering decision ladder

Materials development is already organised as a sequence of increasing

cost and commitment.

A typical progression may move through:

- database search;
- composition-based screening;
- reduced tabular modelling;
- higher-fidelity electronic-structure calculation;
- thermodynamic or kinetic modelling;
- atomistic simulation;
- coupon manufacture;
- physical characterisation;
- component-scale testing;
- qualification.

Each stage consumes more time,

compute,

material,

specialist labour

or equipment access.

The value of a low-cost model is therefore not determined only by its

terminal accuracy.

Its value also depends on whether it helps decide where more expensive

work is worth doing.

Geatomica inserts an assurance step before escalation.

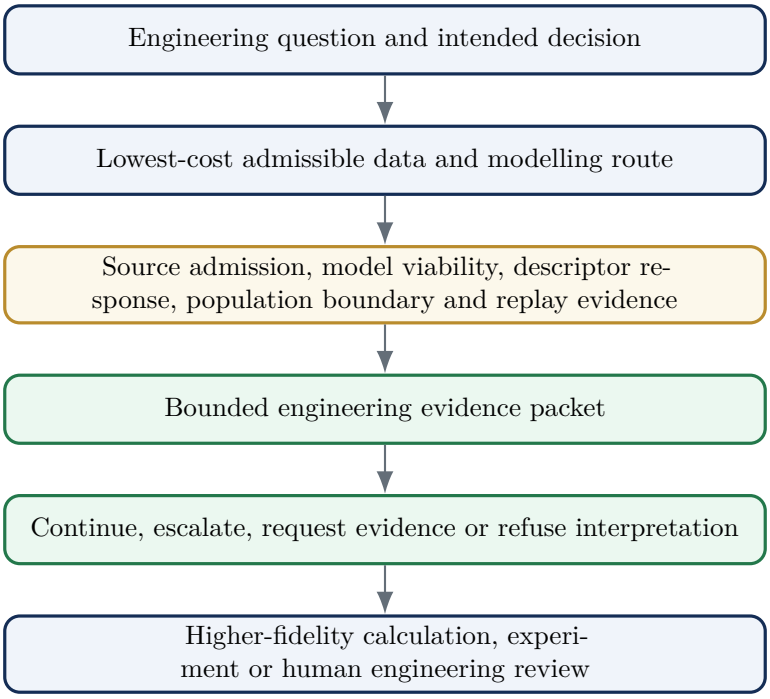


Figure 11.2: The engineering decision ladder. Geatomica sits between a low-cost modelling route and the next expensive action. Its purpose is not to prevent escalation, but to determine when escalation is justified and what uncertainty it must resolve.

The ladder is not a rigid universal workflow.

Different materials programmes begin and end at different stages.

A mature alloy family may already have extensive experimental history.

A novel composition may begin with sparse calculated evidence.

A process-development question may require kinetic or thermal models

rather than a property predictor.

The assurance principle remains the same:

$$\text{use the cheapest route that remains valid for the decision,} \tag{11.1}$$

then escalate when the evidence boundary is reached.

11.6 Continue, escalate, request evidence or refuse

The practical output is not a binary label of

trusted

or

untrusted.

Four bounded outcomes are more useful.

11.6.1 Continue

The current route may continue when:

- the source population is admissible for the intended claim;
- the model crosses both viability doors;
- the material candidate lies within the evidenced population boundary;
- the relevant descriptor account remains coherent under the declared disturbances;
- the result satisfies the required replay contract;
- the requested action remains low consequence.

Continue does not mean universal acceptance.

It means that the current low-cost route remains suitable for the next bounded step.

11.6.2 Escalate

Escalation is appropriate when useful signal exists,

but the current representation cannot support the requested decision.

Examples include:

- row-random performance remains strong,
  but the candidate lies in an unseen chemical system;
- bulk-modulus prediction remains coherent,
  but shear behaviour is weak and directionality matters;
- composition descriptors perform well,
  but the candidate's structure is uncertain;
- descriptor importance and retrained-removal consequence disagree;
- the model is viable,
  but the intended action carries high consequence;
- the reduced route cannot resolve competing physical interpretations.

Escalation may involve:

- a richer structural representation;
- graph or spatial modelling;
- additional DFT calculations;
- CALPHAD or kinetic modelling;

- molecular dynamics;
- finite-element analysis;
- physical characterisation;
- targeted mechanical testing.

11.6.3 Request additional evidence

Some routes should pause before either continuing or escalating.

This occurs when the missing object is not a richer model,

but a clearer source.

Examples include:

- uncertain units;
- missing method provenance;
- unresolved sign conventions;
- unclear tensor-to-scalar reduction;
- ambiguous material identity;
- duplicate or conflicting records;
- missing process history;
- incomplete environmental conditions.

A more complex learner cannot resolve a missing scientific definition.

11.6.4 Refuse interpretation

Refusal is appropriate when:

- the source is inadmissible for the requested claim;
- the model fails both viability doors;
- the candidate is materially outside the evidenced domain;
- the requested action exceeds the claim ceiling;
- the execution or replay evidence is incomplete;
- the result depends on a known leakage route;
- a failed gate is being repaired only through narrative reinterpretation.

Refusal does not mean that the material is unsuitable.

It means that the present evidence cannot justify the requested

conclusion.

Table 11.3: Engineering actions associated with bounded evidence states.

Action	Evidence condition	Typical next step
Continue	Admissible source, viable model, bounded transfer and adequate replay for a low-consequence action.	Retain the reduced route for screening or exploratory interpretation.
Escalate	Useful signal exists, but representation, domain or consequence exceeds the present evidence.	Run higher-fidelity simulation, collect richer descriptors or perform a targeted experiment.
Request evidence	The main uncertainty lies in source meaning, provenance or missing context.	Resolve units, method, identity, tensor meaning or process history before further modelling.
Refuse	An earlier gate fails or the requested claim exceeds the evidence ceiling.	Do not promote the model result into engineering interpretation.

11.7 The bounded engineering evidence packet

A conventional machine-learning result often ends with:

- one performance score;
- one ranked feature list;
- one selected model.

An engineer needs a different object.

The Geatomica evidence packet is intended to include:

1. **Decision statement**

   What engineering question is being asked?

2. **Source statement**

   Which source,

   version,

   population

   and admission rules were used?

3. **Model statement**

   Which representation,

   learner,

   split

   and preprocessing route produced the prediction?

4. **Viability statement**

   Did the model cross both the absolute and relative evidence doors?

5. **Population boundary**

   Which compositions,

   chemical systems,

   structures

   or operating conditions are covered?

6. **Descriptor response**

   Which relationships recur,

   which are fragile

   and which are replaceable?

7. **Replay state**

   Which plan,

   semantic,

   numerical

   and byte contracts passed?

8. **Known defects and unresolved assumptions**

   What remains uncertain or held?

9. **Permitted action**

   What may the engineer do next?

10. **Prohibited interpretation**

    What must not be inferred from the result?

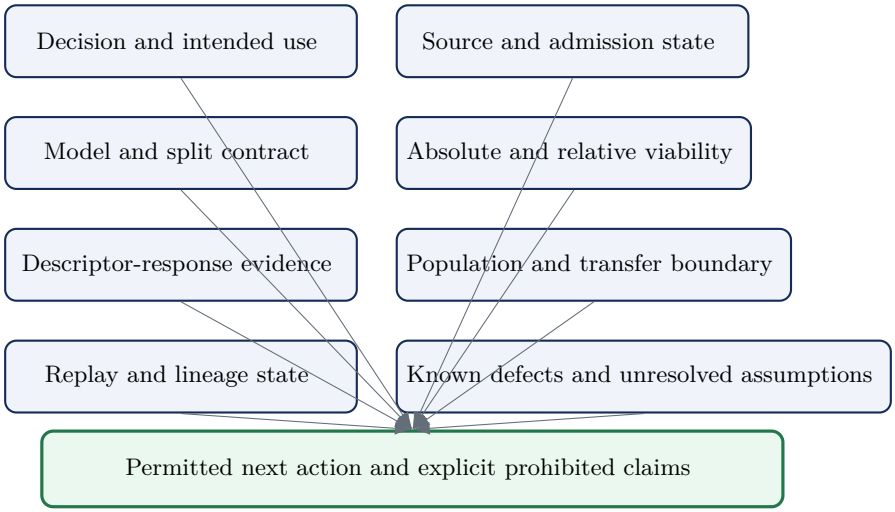


Figure 11.3: Conceptual bounded engineering evidence packet.

11.8 Bounded screening of elastic response

Consider a hypothetical materials-screening task. An engineer wants to identify candidates likely to retain useful bulk stiffness within a defined chemical family.

A conventional route may:

1. train a property model;
2. rank candidate materials;
3. select the top-scoring compounds.

A Geatomica route asks additional questions.

11.8.1 Source admission

Are the elastic targets physically interpretable under the available units, method and bound definitions? If not, the route stops before fitting authority is granted.

11.8.2 Model viability

Does the model satisfy $R^2 > 0$ and $S > 0$? If only the null-separation door passes, the result may support diagnosis but not screening.

11.8.3 Population boundary

Does the candidate lie within a represented composition or chemical system? A row-random score cannot authorise unseen-chemistry use.

11.8.4 Descriptor response

Do density and volume-per-atom relationships recur across models and disturbances? Are they difficult to replace, or do correlated descriptors reconstruct the route?

11.8.5 Engineering action

A candidate may then be retained for low-cost preliminary screening, escalated to a structural or graph model, nominated for a fresh DFT calculation, held pending source resolution, or refused if outside the evidenced domain.

The output is not: "This material has a high predicted modulus." It is closer to: "This candidate ranks favourably within the declared chemical region under a viable bulk-modulus model. Packing-related descriptors recur across the tested routes, but the candidate lies near the composition-domain boundary. Use the result to nominate a higher-fidelity calculation. Do not use it to remove mechanical testing or claim unseen-system transfer."

11.9 Cost-aware materials development

The direct value of materials assurance lies partly in avoiding misallocated effort. A low-cost audit may identify that a candidate lies outside the training domain, the selected target is inadmissible, the model is better than a null but absolutely unusable, a feature interpretation is replaceable, a result fails replay, or a supposedly reduced descriptor set loses critical behaviour.

Each early detection can prevent more expensive work from being commissioned on a weak foundation. However, the monograph does not yet report a controlled industrial cost-saving trial. Any financial example remains illustrative until measured against a real engineering programme.

Cost claim boundary

The present evidence supports the proposition that earlier refusal and better escalation can reduce wasted work in principle. It does not yet quantify industrial savings, return on investment or avoided experimental cost.

11.10 Relationship to established materials tools

Geatomica does not replace density-functional theory, CALPHAD, molecular dynamics, phase-field modelling, finite-element analysis, microscopy, diffraction, mechanical testing or qualification procedures. These methods answer different questions.

Geatomica helps decide which route is justified, which candidates deserve escalation, which assumptions require resolution, which evidence may be used only exploratorily, and which interpretation must stop. It is an assurance layer around the transition from low-cost evidence to higher-cost engineering action.

11.11 Human authority

The final decision remains human. This is not only a safety statement; it follows from the scientific structure of the problem. The model does not possess complete process history, full economic context, manufacturing constraints, regulatory obligations, test availability, programme priorities, or responsibility for the consequence.

The system can structure evidence and limit unsupported promotion. It cannot replace the engineer who holds the complete decision context.

Governing rule

Geatomica may nominate, bound, escalate, hold or refuse. It does not own the engineering decision.

11.12 Chapter conclusion

The practical contribution of Geatomica is not a new material property prediction by itself. It is a decision structure around prediction. The audit determines whether the source can be interpreted, whether the model is viable, whether the explanation persists, where the population boundary lies, whether the result can be replayed, and what action the evidence supports.

This converts model evaluation into an engineering workflow. The next question is whether the evidence packet itself can be trusted to survive long execution, interruption, comparison and later inspection. That requires runtime, recovery and lineage.

Chapter 12

Runtime, Recovery, Scientific Memory and Evidence Lineage

12.1 Why runtime is evidence

Long local runs expose interruption, partial-output and identity risks. A result is not made scientific merely by completing; its declared units, inputs, continuation lineage and terminal state must remain inspectable.

The runtime can therefore be understood as an evidence-state machine rather than primarily as a scheduler: it records which scientific state an execution has actually earned. Execution records what happened, replay adjudicates what repeated, scientific evidence memory records the accepted claim ceiling, and human authority controls downstream action.

12.2 Measured workstation executions

The principal workstation used an AMD Ryzen 7 5800X, 32 GB RAM and an installed NVIDIA RTX 3060. The reported tabular assurance routes were CPU-based and did not require that GPU. JARVIS required approximately 9.93 hours; Matbench required 19.16 hours fresh and 20.60 hours repeat using two bounded CPU workers; Spatial V0.5 required 9.64 and 9.10 minutes; Spatial V0.6.1 committed-unit spans were 20.67 and 16.20 hours. The last pair are not full launcher-to-terminal wall times.

12.3 Recovery is not replay

Atomic units, append-only journals, content hashes, checkpoints and immutable execution identity make interruption recoverable. Resume continues the same identity from verified complete units; replay performs a separately identified execution and compares declared equivalence classes. Partial output is rejected and cannot become terminal evidence.

12.4 Four kinds of memory

Source memory preserves source identity, cohort and provenance. **Execution memory** preserves plans, journals, checkpoints, receipts and lineage. **Scientific evidence memory** preserves measurements, controls, replay states, held defects and bounded conclusions. **Narrative or working memory** preserves summaries, conversations, hypotheses, manuscript prose and AI interpretation.

Memory authority

Narrative memory may help reconstruct the work. It cannot overwrite scientific evidence memory.

12.5 AI detachment and authority

AI and coding agents assisted implementation, critique, diagnosis and documentation. Frozen execution used native scientific software for data, fitting, controls, metrics, receipts and replay. Optional intelligence could propose or explain; it could not promote evidence or grant engineering permission. AI-assisted development is not AI-dependent scientific execution.

12.6 Domain portability boundary

Geatomica is the demonstrated materials implementation. Atomic unit identity, recovery, replay, memory separation and authority transitions operate on declared scientific objects and are portable at the control-logic level. Property semantics, admission rules, thresholds, models and engineering consequences remain domain-specific. Cross-domain scientific validity remains untested.

12.7 Publicly evidenced programme scale

A scientific unit is the smallest contract-bound piece of a declared experiment that the runtime is allowed to call complete on its own. It binds the scientific coordinates, expected outputs, and execution identity. It commits only after required outputs exist, identities and schemas validate, and the commit completes atomically. A unit is not a fixed amount of computation: it may contain one estimator fit, several fits, target-permutation controls, a representation comparison, or a solver/scenario calculation. Unit counts therefore measure accepted scientific boundaries, not FLOPs, CPU-hours, or model fits.

Route	Fresh	Repeat/ replay	Unit executions	Additional estimator or comparison work	Time evidence
JARVIS	48	48	96	28,248 numeric cells; 192 file pairs	approximately 9.93 h fresh
Materials Project	64	64	128	37,924 cells; 916 strict elastic exceedances	total comparable time unresolved
Matbench V0.2	20	0	20	one baseline + five null fits per unit; 120 fits	29 min 40 s elapsed
Matbench V0.3	3,440	3,440	6,880	6,880 fits; 5,680 individual and 1,160 group removal refits across both identities; 65,740 cells	19.16 / 20.60 h
Spatial V0.5	256	256	512	byte, semantic, numeric identity PASS	9.64 / 9.10 min
Spatial V0.6.1	576	576	1,152	1,152 Ridge fits; 864 null fits; 1,728 output pairs; 927,361 cells	20.67 / 16.20 h committed spans

Other sealed spatial predecessors	28	20	48	V0.1–V0.4 retained as distinct routes	route-specific receipts
Totals	**4,432**	**4,404**	**8,836**	Unlike objects are not summed	—

The 4,404 later executions comprise 3,440 same-environment repeat units and 964 predecessor-linked replay units. At least 1,059,273 explicitly counted numeric fresh-to-repeat/replay comparison cells were adjudicated across routes exposing comparable counts. This is not the total number of calculations performed. Approximately 230,370 provenance-separated JARVIS plus Materials Project source records were touched (75,993 plus approximately 154,377). Matbench contributes 260,839 task-record instances, not unique materials and not an additive source census.

12.8 Scientific-unit anatomy

Matbench V0.2 gives the simplest distinction between units and fits: 20 official-fold units each contained one baseline fit and five target-permutation null fits. Thus **20 scientific units represented 120 estimator fits**. In Matbench V0.3, each of 3,440 task–fold–operation units contained a complete fit with fold-local imputation and scaling: 20 baseline/perturbation, 2,840 individual removal, and 580 group removal units. Spatial V0.6.1 bound target, population view, fold, representation, and target state: 144 valid plus 432 paired-null units per execution.

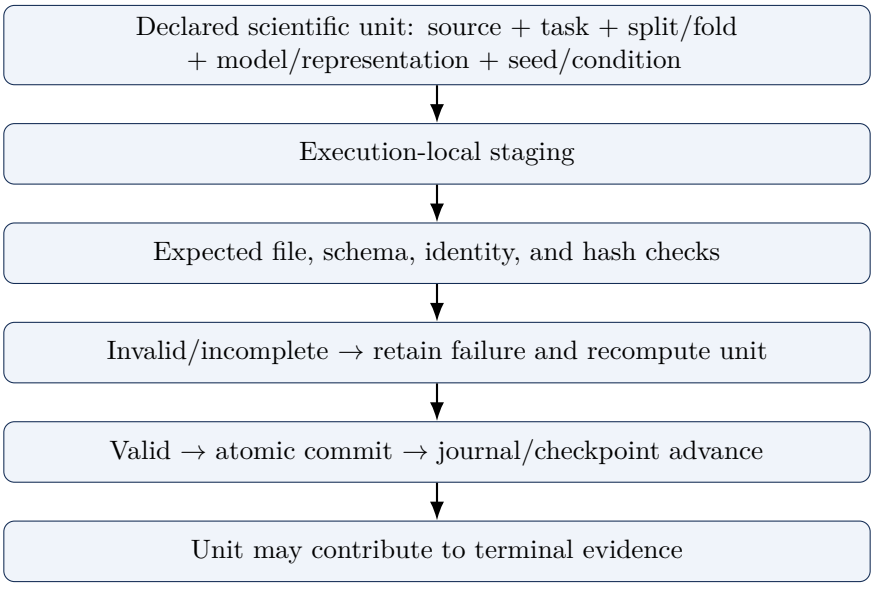


Figure 12.1: Only complete contract-valid units advance accepted execution state. A visible partial file is not evidence that the scientific unit completed.

Verified continuation

Resume from the last verified scientific boundary, not from the last visible file.

12.9 Observed continuity and recovery

The JARVIS replay was interrupted and resumed under the same execution identity; an incomplete staging unit was rejected and recomputed rather than merged or duplicated. The Materials Project replay also used a later resume launch while retaining only verified complete units. Separately, a controlled-fixture recovery matrix terminated real child processes at 11 declared fault boundaries and recovered each fixture to three unique committed units. These are controller fault-injection results, not 11 scientific experiments. A reported JARVIS zero-commit launch-envelope failure was not promoted into this measured register because a definitive run-local receipt was not isolated.

Fresh means a new identity with no predecessor work reused. Resume continues the same identity from verified commits. Repeat is a separately identified execution under the declared same environment where applicable. Replay is a new predecessor-linked identity compared under declared contracts. **Resume is not replication. Replay is not automatically external replication.**

12.10 Calendar time, active execution, and a shared workstation

The principal workstation was a shared personal machine rather than a dedicated compute node. Jobs were scheduled around ordinary availability, so calendar duration is much longer than active execution spans. The workstation remained available for ordinary use during parts of the programme; background load was uncontrolled and is not a performance benchmark.

Elapsed run time is launcher/process duration where a terminal receipt supports it. Summed unit wall time adds unit durations and can exceed elapsed time under parallelism. Committed-unit span runs from first to final accepted commit and can exclude setup or packaging. Calendar elapsed includes review, interruption, ordinary use, and unknown gaps and must not be called runtime. Spatial V0.6.1's 20.67 h fresh and 16.20 h replay values are committed-unit spans.

```
[RUN] - [ordinary use/review] - [RUN] - [interrupt/checkpoint] - [unavailable] -
                        [RESUME/RUN] - [review] - [REPLAY]
```

Scientific identity follows committed evidence, not continuous ownership of the computer.

12.11 Replay does not update accepted science by itself

Execution completion may update execution memory. It must not automatically upgrade the scientific claim. A chronologically separate post-replay demonstration first submitted an eligible claim without human confirmation: the gate refused it and wrote zero accepted scientific-memory objects. Explicit founder adjudication then wrote and read back exactly two bounded local claims: Matbench fitted-use versus retrained-removal disagreement under same-environment repeat, and Spatial bulk metric geometry under exact cross-view local replay. Engineering action remained NONE.

Attempts to strengthen Spatial shear polarity and Materials Project elasticity beyond their ceilings were refused with zero positive promotions. Spatial shear remained conditional because replay exactly reproduced the cross-view failure; Materials Project elasticity retained its mixed replay and admission HOLD.

EXECUTION → COMPARISON → MACHINE CANDIDATE → HUMAN ADJUDICATION → BOUNDED MEMORY UPDATE OR HOLD → READ-BACK

The human controls the evidence-state transition, not merely manuscript wording. A scientific-memory update is not fabrication, deployment, qualification, or regulatory permission.

12.12 What remains unmeasured

The programme does not establish universal efficiency, production throughput, energy efficiency, external replication or human-labour savings. The source providers' DFT corpora were not recomputed locally.

Chapter 13

Governance and Claim Authority

13.1 Why governance appears last

Governance is introduced after the materials experiments, engineering decisions and execution evidence because it is the response to a problem revealed by the experiments.

The programme produced states such as high prediction with low retrained-removal consequence, positive null separation with negative absolute performance, source availability with unresolved admissibility, and numerical replay with byte divergence. Governance preserves the state of each evidence object and controls what may be claimed.

Governing rule

Governance does not create scientific evidence. It prevents evidence from being promoted beyond the state it has actually reached.

13.2 The evidence-to-authority chain

The chain from public data to engineering authority is:

$$\text{availability} \rightarrow \text{admission} \rightarrow \text{viability} \rightarrow \text{reliability} \rightarrow \text{replay} \rightarrow \text{bounded authority}. \tag{13.1}$$

No later gate repairs an earlier failed requirement.

Table 13.1: Evidence-to-authority gates.

Gate	Operational question	Failure state
Availability	Does the source expose the required field or object?	Record absent or incomplete.
Admission	Are units, method, sign and semantics adequate?	Source hold.
Viability	Does the model pass absolute ($R^2 > 0$) and relative ($S > 0$) doors?	Hold or refuse.
Reliability	Does descriptor explanation survive disturbances and retrained removal?	Bounded use or refusal.
Replay	Does execution satisfy required replay contracts?	Replay hold.
Authority	What action does the surviving evidence permit?	Bounded permission only.

13.3 Allowed terminal evidence states

The monograph uses four principal terminal states: **[ACCEPTED]**, **[BOUNDED]**, **[HOLD]**, and **[REFUSE]**.

13.4 Chapter conclusion

Governance preserves the distinctions among data, evidence, interpretation, permission and authority. A result may influence an engineering decision only to the extent that the required evidence gates have survived.

Chapter 14

Discussion, Novelty and Closest Prior Work

14.1 The central result

The central result of this monograph is the experimental separation of several evidence objects that are often allowed to collapse into one another:

$$\begin{aligned} &\text{prediction} \neq \text{viability}, && \text{fitted use} \neq \text{retrained dependence}, \\ &\text{availability} \neq \text{admissibility}, && \text{repetition} \neq \text{transfer}. \end{aligned} \tag{14.1}$$

14.2 Distinguishing three contribution classes

To evaluate the contribution of this work, it is necessary to distinguish three distinct contribution classes:

1. **Scientific Contributions**: The conceptual separation of descriptor authority from fitted importance, the two-door model viability rule ($R^2 > 0$ and $S > 0$), and source admission requirements prior to property fitting.
2. **Empirical and Failure-Witness Contributions**: The complete audit and replay of JARVIS-DFT (75,993 records), the adverse elasticity result in the Materials Project (154,377 records), the demonstration of the null-separation paradox, and the sealed 3,440-unit Matbench removal programme and separately identified 3,440-unit exact same-environment repeat.
3. **Research-Instrument Contributions**: The construction of atomic contract-bound scientific units, deterministic replay comparators, verified checkpoint recovery, and claim-addressable evidence lineage.

14.3 Closest prior method families

Geatomica overlaps with and builds upon several established scientific fields:

Adjacent methods often address different evidence objects. The contribution here is the explicit separation and interaction of those objects under one bounded scientific programme. Table 14.1 summarises primary concerns; it does not assert that every implementation in an adjacent family is limited to one function.

Table 14.1: Closest method families and the evidence object kept separate in Geatomica.

Method family	Primary object reused	Object kept separately inspectable
XAI and attribution	Fitted feature use and local/global attribution	Retrained dependence and bounded mechanism claims
Feature-selection stability	Selection recurrence under resampling	Prediction viability and engineering permission
Uncertainty quantification	Predictive uncertainty or confidence	Source admission and evidence authority
Applicability domain and OOD	Population and transfer boundary	Semantic admission and downstream action
Benchmark suites	Standard tasks, splits and metrics	Claim ceilings when evidence objects disagree
FAIR and provenance [16, 17]	Traceability, stewardship and lineage	Whether traced evidence has earned admission or action
Research objects and workflow systems [18, 19, 20]	Packaging, execution and reconstruction	Resume/replay meaning and accepted scientific state
Engineering assurance	Structured argument and failure containment	Empirical scientific gate results
Model governance [21]	Lifecycle and organisational risk	Scientific evidence state before organisational promotion

14.3.1 Feature importance and attribution

Attribution algorithms such as SHAP [5], LIME [6], and Gini/permutation importance rank feature contributions within a fitted model. Geatomica demonstrates that fitted feature attributions do not establish retrained dependence or physical necessity. By placing fitted attributions beside retrained removal and multi-axis disturbance, Geatomica distinguishes features that a specific model uses from features that are associated with retrained-removal consequence under the declared representation.

14.3.2 Feature-selection stability

Stability selection [7] evaluates the consistency of selected feature sets across subsamples. Geatomica incorporates stability analysis into a broader materials-assurance programme by measuring explanatory persistence under noise injection, population splitting, and cross-source transfer boundaries.

14.3.3 Applicability domains and out-of-distribution robustness

Quantitative structure-property relationship (QSPR) applicability domains [8] define the chemical space where model predictions remain valid. Geatomica operationalizes population boundaries for both predictive accuracy and explanatory stability, refusing to authorize engineering actions when candidates cross structural or compositional boundaries.

14.3.4 Uncertainty Quantification (UQ)

Uncertainty quantification methods [9] estimate prediction variance and model confidence. Geatomica complements UQ by measuring evidence sufficiency—determining whether an uncertainty bound permits specific engineering actions (such as screening, escalation, or physical testing).

14.3.5 Materials benchmark suites

High-throughput materials repositories such as AFLOW [10], OQMD [11], and Matbench [3] provide standardized evaluation datasets. Geatomica operates inside these benchmark datasets while demonstrating that high benchmark scores cannot repair underlying source admission failures or unverified target definitions.

14.3.6 Workflow provenance and scientific reproducibility

Reproducible research standards [14, 15, 16] emphasize computational provenance. Geatomica connects cryptographic execution receipts directly to manuscript claim ceilings, enforcing fail-closed decision rules when execution logs are incomplete or non-reproducible.

14.3.7 Engineering software assurance

In contrast to purely predictive materials ML models [22, 23], Geatomica draws upon safety-case and software-assurance principles [24, 25, 26]. It is designed as an assurance layer that bounds model authority before predictions reach engineering workflows.

14.4 The novelty claim

The novelty claim of Geatomica is combinational: it brings together source admission, absolute/relative viability, descriptor-response testing, retrained removal, multi-contract replay, and bounded engineering permission into a single fail-closed architecture.

The novelty claim concerns the composition and explicit separation of these evidence objects, not the invention of each constituent method family.

14.5 Chapter conclusion

The contribution lies in ensuring no single evidence object can impersonate another when a materials model begins to influence engineering decisions.

Chapter 15

Limitations, Threats to Validity and Future Falsification Programme

15.1 Limitations define the present result

Geatomica has demonstrated a materials-assurance method across large public datasets, multiple model families, and governed replays. It has not yet demonstrated physical causal discovery, experimental transfer, or universal descriptor hierarchy.

15.2 Falsification programme

Future work should prioritize 10 targeted falsification experiments: matched full-feature sham refits; alternative model families for the removal experiment; direct grouped or conditional fitted perturbation; tensor- and method-resolved Materials Project elasticity reconstruction; external clean-room reproduction; cross-platform replay; physical materials validation; a prospective industrial decision trial; authority-threshold sensitivity; and the first genuine non-materials-domain instantiation of the portable control logic.

15.3 Chapter conclusion

The strongest next work is not internal expansion, but an attempt to break current claims through new sources, spatial models, and physical measurements.

Conclusion

Materials engineers rarely receive the whole process; they receive what remains. A microstructure, a fracture surface, a calculated database record, or a trained machine-learning model is a surviving structure.

Geatomica emerged from the empirical demonstration that prediction is not viability, importance is not necessity, availability is not admissibility, and replay is not universal truth. Its purpose is not to declare every model trustworthy, but to measure the boundary of trust and issue a smaller, clearer, and more defensible permission to act.

If you remember one thing: no computational evidence object should be allowed to stand in for a stronger one that has not actually been tested.

Chapter A

Experiment Register

Route	Scale	Replay state	Principal outcome
JARVIS tabular	48 + 48 units	28,248 cells; max delta $\sim 1.07 \times 10^{-14}$	viable carrier with target/model/population dependence
Materials Project	64 + 64 units	mixed; 916 strict exceedances, all elastic	admission and viability holds retained
Matbench removal	3,440 + 3,440 units	65,740 cells exact	fitted importance diverged from retrained removal
Spatial persistence	256 materials	sealed predecessor route	six-bin angle advantage failed to persist
Spatial V0.5 DOE	256 + 256 units	byte/metadata/numeric PASS	metric geometry strongest compact block
Spatial V0.6.1	576 + 576 units	1,728 byte-identical pairs; 927,361 cells exact	bulk geometry confirmed; shear polarity conditional

Chapter B

Claim-to-Evidence Matrix

Table B.1: Claim-to-evidence matrix.

Claim	State	Boundary
JARVIS non-null predictive structure	Supported	Within frozen JARVIS populations and representations.
JARVIS numerical replayability	Supported	Local locked replay at 10^{-10} tolerance.
Target-specific descriptor enhancement	Supported	Within frozen non-leakage register.
MP formation energy	Bounded support	Population-level recurrence across sources.
MP raw elasticity physical use	Refused	Extreme and negative values triggered scrutiny; unresolved tensor-, method-, and convention-level evidence prevented the strongest physical-use category under the declared admission rubric.
Importance proves necessity	Not authorised	Matbench establishes an exactly repeated measurement disagreement, not physical necessity or causal irrelevance.

Chapter C

Reproducibility, Runtime and Hardware Register

Route	Runtime	Comparison	State
JARVIS	∼9.93 h fresh	28,248 cells; 60/192 raw files byte-divergent	numerical PASS within frozen tolerance
Materials Project	not recovered	37,924 cells; 916 exceedances	mixed replay; elastic hold retained
Matbench	19.16 / 20.60 h	65,740 cells	exact numerical and semantic identity
Spatial V0.5	9.64 / 9.10 min	256 + 256 units	byte, metadata and numerical identity
Spatial V0.6.1	20.67 / 16.20 h committed spans	1,728 output pairs; 927,361 cells	byte, metadata and numerical identity; max delta 0

The principal workstation was an AMD Ryzen 7 5800X with 32 GB RAM and an installed RTX 3060. The reported tabular routes were CPU-based. Same-environment replay is not external replication.

Chapter D

Scientific Reproduction Configuration Register

This appendix consolidates the claim-relevant scientific settings. It is a method register, not a release of the private execution or authority machinery.

JARVIS-DFT audit

Item	Frozen scientific configuration
Source and cohorts	JARVIS-DFT `dft_3d`; source census 75,993; four-property common cohort 23,824; target-specific admissible cohorts retained separately.
Targets	Formation energy per atom, band gap, bulk modulus K_V, shear modulus G_V.
Representations	Common leakage-restricted descriptor lane and target-specific lane under the frozen protocol.
Models	Ridge, Random Forest, Extra Trees and HistGradientBoosting under the frozen source protocol.
Population and scale	Seeds 11, 23, 37; row-random, composition-group and chemical-system splits; 2k, 5k, 10k and full-data scales.
Controls	Noise, dropout, feature shuffle, descriptor removal and paired target-permutation nulls.
Replay	48 fresh + 48 replay units; 28,248 compared numeric cells; maximum absolute difference approximately 1.07×10^{-14}.

Materials Project audit

Item	Frozen scientific configuration
Source	Materials Project checkpoint dated 28 July 2026; approximately 154,377 unique source records. Elastic route: 13,028 source records; 13,021 common-cohort records.
Targets	Formation energy, band gap, bulk modulus and shear modulus; elastic viability analysis keeps bulk and shear separate.
Representations	Eleven common descriptors and sixteen target-specific descriptors.

Preprocessing	All four model families used the same fold-local preprocessing: numeric coercion and median imputation followed by StandardScaler; categorical crystal-system values received most-frequent imputation and dense one-hot encoding with unknown categories ignored. All transformers were fitted on the training partition only.
Models	Ridge ($\alpha = 1$); Random Forest (200 trees, minimum leaf 2); Extra Trees (300 trees, minimum leaf 2); HistGradientBoosting (250 iterations, depth 6, learning rate 0.05).
Population and scale	Seeds 11, 23, 37; 80/20 row-random, reduced-composition-group and chemical-system-group splits; 2k, 5k, 10k and full scales.
Null	At full scale, each eligible valid cell was paired with the mean R^2 across 30 target-permutation refits. Training targets were permuted while the held-out targets and other declared coordinates remained fixed.
Metrics and replay	`sklearn.metrics.r2_score` and MAE. 64 fresh + 64 replay units; 37,924 compared numeric cells; 916 strict tolerance exceedances, all elastic.

Matbench importance/removal audit

Item	Frozen scientific configuration
Tasks and size	`matbench_mp_e_form` (132,752), `matbench_mp_gap` (106,113), `matbench_log_kvrh` (10,987), and `matbench_log_gvrh` (10,987); official five folds.
Representation	142 Magpie composition descriptors. Fold-local median imputation and StandardScaler fitted on training data only.
Model	scikit-learn 1.8.0 MLPRegressor; hidden layers 128/64; ReLU; Adam; $\alpha = 10^{-4}$; initial learning rate 10^{-3}; maximum 300 iterations; early stopping; validation fraction 0.15; patience 20; random state 11.
Fitted importance	Held-out permutation importance, five repeats, fixed random state; fold-level aggregation.
Removal	Complete refitting after each individual descriptor removal and after each of 29 declared group removals; preprocessing refitted within the changed representation.
Comparison	Spearman rank association, top-10 overlap, MAE change and target-level group summaries.
Repeat	3,440 fresh + 3,440 repeat units; two bounded CPU workers; 65,740 compared numeric cells; maximum difference 0.

Spatial V0.5 representation DOE

Item	Frozen scientific configuration

Cohort and targets	Frozen 256-record JARVIS cohort; bulk modulus K_V and shear modulus G_V; contiguous JID-ordered four-fold split.
Predictor	`sklearn.linear_model.Ridge(alpha=1e-6)`; seed 20260809.
Baseline	Mean, standard deviation, minimum and maximum atomic number plus mean and standard deviation atomic mass.
Factors	A: symmetry/regime; B: coordination topology; C: continuous metric geometry; D: repaired six-bin angular geometry; E: electronegativity/polarity proxies.
Design and controls	Full 2^5 factorial, 32 configurations, 256 scientific units per execution; translation, rotation, atom-permutation and supercell controls; tolerance 10^{-8}.
Replay	Required; 256 fresh + 256 replay units; byte, semantic and numerical identity passed.

Spatial V0.6.1 confirmation

Item	Frozen scientific configuration
Cohort and model	Same frozen 256-record cohort and Ridge($\alpha = 10^{-6}$).
Population views	JID-order contiguous four-fold view and chemical-system-group four-fold view; decisions kept separate.
Configurations	BASE, B, E, B+E, C, C+E and dimension-matched sham configurations defined before execution.
Null and shams	Three paired-null seeds (20260809–20260811); training target only permuted. Deterministic target-blind within-partition feature shams; no train/test movement.
Decision rules	Bulk C required $\Delta R^2 \geq 0.05$ against BASE and SHAM_C in each view with at least 3/4 positive folds. Shear E-over-C required $\Delta R^2 \geq 0.02$ against C and C+SHAM_E in each view with at least 3/4 positive folds.
Controls and replay	Representation tolerance 10^{-8}; same-environment replay tolerance 10^{-10}; 576-unit ordered plan; 576 fresh + 576 replay; 1,728 output pairs; 927,361 numeric cells; maximum replay difference 0.

Unresolved reproduction fields. Exact provider-side generation settings for third-party source corpora remain governed by the source providers. Public reproduction begins from the cited source datasets and frozen cohort/split identities; it does not reproduce the providers' underlying first-principles calculations.

Chapter E

Public Evidence Package

Release condition

No figure, table or numerical headline remains in the public monograph unless it can be traced to a machine-readable evidence artefact.

Chapter F

Evidence Verification Statement

F.1 Verified publication state

The public evidence edition was checked against the executable metric implementation, sealed execution and replay receipts, accepted scientific adjudications and the final Spatial V0.6.1 assessment. The reported R^2 definition follows the evaluated-set target mean used by `sklearn.metrics.r2_score`. Matbench fresh/repeat evidence, Materials Project same-denominator viability results and Spatial V0.6.1 fresh/replay states were reconciled across the monograph and companion papers.

The verification establishes internal consistency and same-environment replay states. It does not constitute peer review, external replication, physical qualification, regulatory approval or authority for 512-material scaling.

Declarations

Funding. This research was conducted independently without dedicated external research grant funding.

Competing interests. The author is founder of a venture developing software and services related to scientific evidence assurance and engineering decision support described in this work.

Data and code availability. The underlying source datasets are available through JARVIS-DFT, the Materials Project and Matbench under their respective access and licence terms. The maintained public research compendium is available at github.com/JosephMaxwell02/materials-ml-failure-analysis; its archived v1.0.0 release is doi:10.5281/zenodo.21919563. Compendium scripts reconstruct readouts from accepted derived tables. A separate scientific verification rerunner, archived at doi:10.5281/zenodo.21939447, performs selected bounded fitting calculations; it does not rerun every published campaign. Exact historical Materials Project inputs are not redistributed.

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AI assistance. AI and coding tools assisted code development, critique, diagnosis, evidence organisation, and manuscript preparation. They did not generate source target data. Frozen analyses were executed by conventional scientific software under declared protocols, and scientific interpretations and conclusions were reviewed by the author.

Author contribution. Joseph Maxwell is the sole human author and research lead of the reported programme.

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